

HoneyPot Deployment and Threat Analysis Using Cowrie and Wazuh SIEM

EXECUTIVE SUMMARY

This project presents the design, deployment, and evaluation of a secure honeypot environment for monitoring attacker behaviour within an isolated virtual laboratory. A **Cowrie SSH honeypot** was deployed on an Ubuntu Server using Docker to emulate a vulnerable system capable of attracting and recording malicious activity. **Kali Linux** was used to simulate common SSH-based attacks, while **Wazuh SIEM** collected, centralized, and analyzed security logs in real time.

The primary objective of the project was to investigate how attackers interact with exposed SSH services, understand post-compromise behaviour, and demonstrate how Security Information and Event Management (SIEM) solutions can enhance threat detection and incident analysis. Custom detection rules were developed in Wazuh to identify suspicious activities such as repeated authentication failures, command execution, and file download attempts. The captured events were visualized through the Wazuh dashboard, providing clear insight into attack patterns and system activity.

To further enrich the investigation, selected honeypot logs were analysed using an AI-assisted approach and mapped to the **MITRE ATT&CK Framework**. This supported the identification of adversary techniques while demonstrating how AI can complement traditional security monitoring. All AI-generated observations were manually validated against the original logs to ensure accuracy and maintain analytical reliability.

The project successfully demonstrates the value of combining honeypot technology, centralized log management, and threat intelligence frameworks to improve security monitoring and attacker analysis. It also highlights the importance of conducting cybersecurity research within an isolated environment, allowing realistic attack simulations without impacting production systems. The knowledge and practical experience gained through this project provide a strong foundation in deception technologies, SIEM operations, threat detection, and cybersecurity monitoring practices.

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1. HONEYPOT LAB

1.1 VM Setup

A virtual lab was created to safely test the behavior of attackers with the help of VirtualBox. There were three virtual machines used in this setup. The computer used by the attacker was Kali Linux, Cowrie honeypot was running on Ubuntu Server and logs were collected and examined using the Wazuh server. Two network adapters were installed on each machine.

The first adapter provided simple access to the internet via NAT when it was required. The second one used a host-only network where the IP addresses were in the 192.168.56.x range. This setup allowed everything to be contained within the laboratory, and all the machines to communicate. The lab was totally cut off from the actual network as a result of this setup. This type of configuration is used in cybersecurity labs to replicate attacks in a safe way (NIST, 2020). The overall architecture of the lab environment is shown in Figure 26 (Appendix), illustrating the interaction between the attacker, honeypot, and SIEM system.

1.2 Software Tools

The following tools were used:

1. The Cowrie Honeypot

Cowrie was the main honeypot (Cowrie, 2026). It logs the work of attackers, such as commands and attempts to log in, and acts as a fake SSH server.

2. Ubuntu Server

The Cowrie honeypot was operated by Ubuntu. It provided a dependable platform on which to run the system.

3. Docker

Cowrie was deployed and easily used with Docker. It expedited deployment and made setup easier.

4. Wazuh SIEM

Wazuh was used to collect and analyze logs generated by the honeypot (Wazuh, 2026). It also helped in the tracking of the activity of attackers and alerts.

5. Kali Linux

Kali Linux was run on the computer of the attacker. It was used to test the honeypot and perform SSH attacks.

Cowrie was chosen in this project above other tools such as T-Pot (T-Pot, 2026) since it is lightweight, easier to install and is mainly used to attack SSH.

1.3 Network and Gateway Configuration

The lab network was configured using two types of network adapters so as to strike a balance between security and connectivity. When the virtual machines had to connect to the internet such as to install updates or tools, they were able to do so courtesy of utilization of NAT by the first adapter. The second adapter used a host-only network whose IP addresses were in the range 192.168.56.x.

This had enabled all the virtual machines in the lab to communicate with each other. At the same time, the host-only network ensured that all the activity of the attack was contained within the lab and was not emitted

to other systems. As a result, testing could be done in a risk-free, controlled environment. Through this setup, the lab was able to stay isolated, prevent undesirable external interactions, and comply with the relevant ethical security measures. No bridged networking was used in this setup, ensuring that the lab environment remained completely isolated from the host system and external networks.

This setup also made it easier to control and monitor all network activity within the lab environment. It ensured that any attack traffic remained contained and could be safely analysed without any external impact.

1.4 Lab Testing and Isolation Verification

The lab setting was tested to make sure that all was going on as planned. In order to check whether the virtual machines were able to communicate among themselves, ping tests were initially conducted among them. The ability of the attacker system to reach the target was then verified by testing an SSH connection from the Kali machine to the honeypot.

The Cowrie honeypot was then checked to ensure that it was logging even in case of attack. These logs were then verified by the use of the Wazuh system to ensure that they were being collected and presented properly. All these tests confirmed that the set up was working as intended and that no other systems were affected by any activity that was going on in the lab.

2. RESEARCH-BASED HONEYPOT ANALYSIS

2.1 Executive Summary

To observe and understand the actions of attackers in a safe and controlled laboratory environment, a honeypot system was deployed in this project. The system was designed to simulate a real system so that attacker behaviour could be monitored without any risk to actual networks or systems.

The honeypot was able to capture different types of attacker activity, including login attempts, commands entered, and file download actions. This information provided useful insight into how attackers try to gain access and what they do after successfully entering a system. All the collected data was sent to a SIEM platform, where it was stored, monitored, and analysed in real time.

In addition, detection rules were created to identify suspicious activities such as repeated login attempts and unusual commands. This helped in automatically detecting potential attacks and understanding attacker behaviour more clearly.

An AI tool was also used to support the analysis process. It helped in summarising the attacker's actions and mapping them to known attack techniques, making the results easier to interpret. The findings from the AI were also checked manually to ensure accuracy.

Overall, this project helped in gaining practical understanding of how cyber attacks occur, how they can be detected, and how security tools such as honeypots, SIEM systems, and AI can be used together to improve cybersecurity monitoring and analysis (NIST, 2020).

2.2 Methodology

In this project, a controlled attack was carried out to study how an attacker behaves in a real-world scenario. The attack was performed using the Kali Linux machine, which acted as the attacker system, while the Cowrie honeypot running on the Ubuntu server was used as the target.

First, an SSH connection was initiated from the Kali machine to the honeypot using port 2222. This port is used by Cowrie to simulate an SSH service. The attacker then attempted to gain access by trying multiple usernames and passwords, which represents a brute-force attack. This type of attack is commonly used by attackers to break into systems by repeatedly guessing login credentials (MITRE, 2026). After several attempts, a successful login was simulated to observe what actions an attacker might perform after gaining access. Once access was obtained, different commands were executed inside the honeypot environment. These included basic shell commands and attempts to download files, which are typical actions performed by attackers after compromising a system. All of these activities were recorded by the Cowrie honeypot, allowing detailed observation of attacker behaviour.

This approach was chosen because it closely reflects real-world attack techniques. Attackers often begin by attempting to gain access using weak or guessed credentials and then move on to executing commands or downloading tools to further exploit the system. By simulating this behaviour in a controlled lab environment, it was possible to safely study and analyse these actions without any risk to real systems.

All activities were conducted within an isolated virtual lab to ensure that no external systems were affected. This also ensured that the experiment followed proper ethical guidelines and cybersecurity best practices.

2.3 Attack Simulation

To understand how an attacker behaves in a real-life situation, an attack was simulated in this project. The attack was carried out using the Kali Linux machine, which acted as the attacker system, while the Cowrie honeypot was used as the target system.

At the beginning, the attacker attempted to log in by using common usernames such as “root” along with different password combinations. This type of activity is known as a brute-force attack, where an attacker tries multiple combinations repeatedly in order to gain access to the system. This method is commonly used in real-world attacks, especially when systems have weak or easily guessable credentials (MITRE, 2026).

After several login attempts, a successful login was simulated using weak credentials. This step was important to observe what actions an attacker might take after gaining access to a system. Once inside the honeypot, the attacker executed various commands such as *ls*, *pwd*, and *whoami* to explore the system environment. In addition, a file download attempt was made using the *wget* command. These actions represent typical post-exploitation behaviour, where an attacker interacts with the system and attempts to download additional tools or files (Spitzner, 2003).

All these actions, including failed login attempts, successful access, and commands executed, were recorded by the Cowrie honeypot. The honeypot captured detailed logs such as timestamps, IP addresses, usernames, and commands used by the attacker. This provided valuable data for analysing attacker behaviour.

This simulation helped to demonstrate a typical attack sequence: first gaining access through brute-force attempts, then exploring the system, and finally attempting further actions such as downloading files. By analysing this behaviour, it becomes easier to understand how attackers operate and how such activities can be detected and prevented in real-world systems (NIST, 2020).

2.4 Log Collection and SIEM Integration

The logs generated by the Cowrie honeypot were all collected during the experiment, and sent to the Wazuh system to track and assess. The honeypot logs were continuously forwarded to the Wazuh server by the Wazuh agent that was installed on the Ubuntu computer. The logs were automatically processed and arranged once they arrived at Wazuh. The SIEM platform stored the logs and enabled the search and analysis of the attacker activity.

Also, it allowed real-time monitoring and therefore any new attacks or suspicious activity could immediately be viewed on the dashboard. Also, Wazuh provided data visualizations, which facilitated understanding such trends as the number of files downloaded, commands entered, and attempts to log in. The setup adhered to the OWASP Logging Cheat Sheet's recommendations to guarantee appropriate logging procedures (OWASP, 2026). This made sure that significant security incidents were accurately documented and could be utilized for analysis.

The honeypot and SIEM platform's integration made it possible to efficiently and systematically gather, analyze, and comprehend attacker behavior. Moreover, the Wazuh system was extended with a dashboard that shows the activity of an attacker, such as file downloads, executed commands, and attempts to log in. This dashboard has helped to easily understand the attack patterns and monitor the actions of the attacker in real time.

2.5 Detection Rules and Alerts

In order to detect suspicious and malicious activity, custom detection rules were developed in the Wazuh system (Wazuh, 2026). The following rules were developed based on the behavior observed in the honeypot logs. Multiple failed attempts to log in using SSH are evidence of a brute-force attack and they were identified with the help of the first rule.

The other rule was created to determine suspicious commands such as `wget` and `chmod` that are commonly used by hackers after gaining access to a system. Moreover, a rule was created to identify the presence of file download activity which may indicate that the attacker is trying to download harmful tools or files (OWASP, 2026). These alerts enabled the system to automatically recognize and point out attacker activities in real-time. This simplified the process of monitoring the actions of attackers and quickly establishing the type of activity that was going on within the honeypot. These detection rules were based on identifying patterns in log data, such as repeated authentication failures and execution of specific high-risk commands.

2.6 Data Analysis and Findings

The analysis of the logs collected in the honeypot revealed clear trends in attacker activity. There were multiple login attempts from the attacker system, which indicates a brute-force attack where different username and password combinations are repeatedly tried. In some cases, successful logins were also observed, showing that weak credentials can be easily exploited (MITRE, 2026).

Once access was gained, the attacker executed several commands within the honeypot. These included basic shell commands as well as actions related to downloading files. The presence of file download activity suggests that the attacker may have been attempting to bring additional tools or potentially malicious files into the system for further exploitation (OWASP, 2026).

For example, an attacker from the IP address 192.168.56.102 successfully logged into the honeypot and executed commands. This reflects a typical real-world attack pattern, where an attacker first gains access through brute-force techniques and then performs additional actions on the compromised system (Spitzner, 2003).

Further analysis of the logs shows that the attacker behaviour followed a structured sequence: initial access through repeated login attempts, followed by command execution, and then attempts to download files. This step-by-step pattern indicates that the attacker was not only trying to access the system but also preparing to extend control over it.

In addition, the frequency of login attempts and the use of common usernames such as “root” suggest automated or scripted attack behaviour, which is commonly seen in real-world cyber attacks (NIST, 2020). The commands executed inside the honeypot also indicate an attempt to explore the system environment and identify possible vulnerabilities.

Overall, this activity suggests that the attacker was likely trying to maintain access to the system and possibly install or download additional tools to continue the attack. This type of behaviour highlights the importance of monitoring login activity, detecting unusual commands, and preventing the use of weak credentials in real systems.

2.7 AI-Assisted Enrichment and MITRE Mapping

This project analyzed the honeypot logs collected in Wazuh system with ChatGPT, which is an AI tool (OpenAI, 2026).

1. Input

A small set of Cowrie logs was provided to the AI. These logs included the download of the file, successful access and the login attempts of the attacker.

2. Output

Upon the logs analysis, the AI concluded that the attacker had brute-force logins, run commands, and downloaded files after gaining access to the system.

3. MITRE ATT&CK Mapping

The following is how the behavior was mapped to the MITRE ATT&CK framework (MITRE, 2026):

- **T1110 – Brute Force:** A number of attempts to log in were observed.
- **T1059 – Command Execution:** The system carried out commands. □ **T1105 – Ingress Tool Transfer:** The activity of downloading files was evident.

4. Verification

Using the AI results as a comparison with the logs observed in Wazuh, they were verified manually. This verified that the analysis matched the real attacker behavior and was accurate. While the AI provided useful insights, it may not always be fully accurate or context-aware, so manual verification was necessary to ensure correctness. However, AI analysis may lack full contextual understanding and can sometimes produce misleading interpretations, which highlights the importance of human validation.

2.8 Critical Reflection

This project had a number of advantages. It provided an excellent means of understanding the real way attacks are conducted through a honeypot, which was a realistic means of simulation of attacker behavior. Using the Wazuh platform, the system was able to efficiently gather logs and keep an eye on attacker activity. Analyzing the data and identifying suspicious activity was made simpler by the honeypot and SIEM's integration. There were some restrictions, though. Other forms of attacks like web or network based attacks were not addressed in the experiment since it was mainly interested in SSH based attacks. Also, since the lab environment was small it fails to closely replicate a large-scale network in reality.

Ethically, all the activities were performed under isolated and controlled conditions in a laboratory. There were no real systems or external networks that were affected, thus the work was ensured to comply with the ethical and security standards.

Further, some possible improvements to future projects can be identified. To log the different types of attacks, additional honeypots might be included. More complicated attack simulations may be conducted to gain more insight into complex threats. Detection rules and monitoring might also be enhanced to make the system more effective in detecting suspicious activity. In a real-world SOC environment, detection rules may produce false positives and would require continuous tuning and validation. Additionally, in real-world environments, attackers may use more advanced evasion techniques, which may not always be captured by basic honeypot configurations.

2.9 Conclusion

In this project, the honeypot system was effective in capturing and analysing the behaviour of attackers. It recorded login attempts, commands executed, and file download activities, which helped in understanding how an attacker interacts with a system. These activities were clearly monitored and analysed through the integration with the Wazuh SIEM platform, making it easier to identify suspicious behaviour in real time.

The use of an AI tool (ChatGPT) also supported the analysis by helping to explain attacker behaviour and map it to known attack techniques. This made the investigation more structured and easier to understand.

Overall, this project demonstrates that honeypots are a valuable tool in cybersecurity. They provide a safe and controlled environment to study real attack patterns without affecting actual systems. In addition, combining honeypots with SIEM tools and AI-based analysis can improve threat detection, monitoring, and understanding of attacker behaviour. This approach can be very useful in real-world security operations for identifying and responding to cyber threats more effectively.

3. REFERENCES

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- Spitzner, L. (2003) *Honeypots: Tracking Hackers*. Boston: Addison-Wesley.

4. APPENDIX

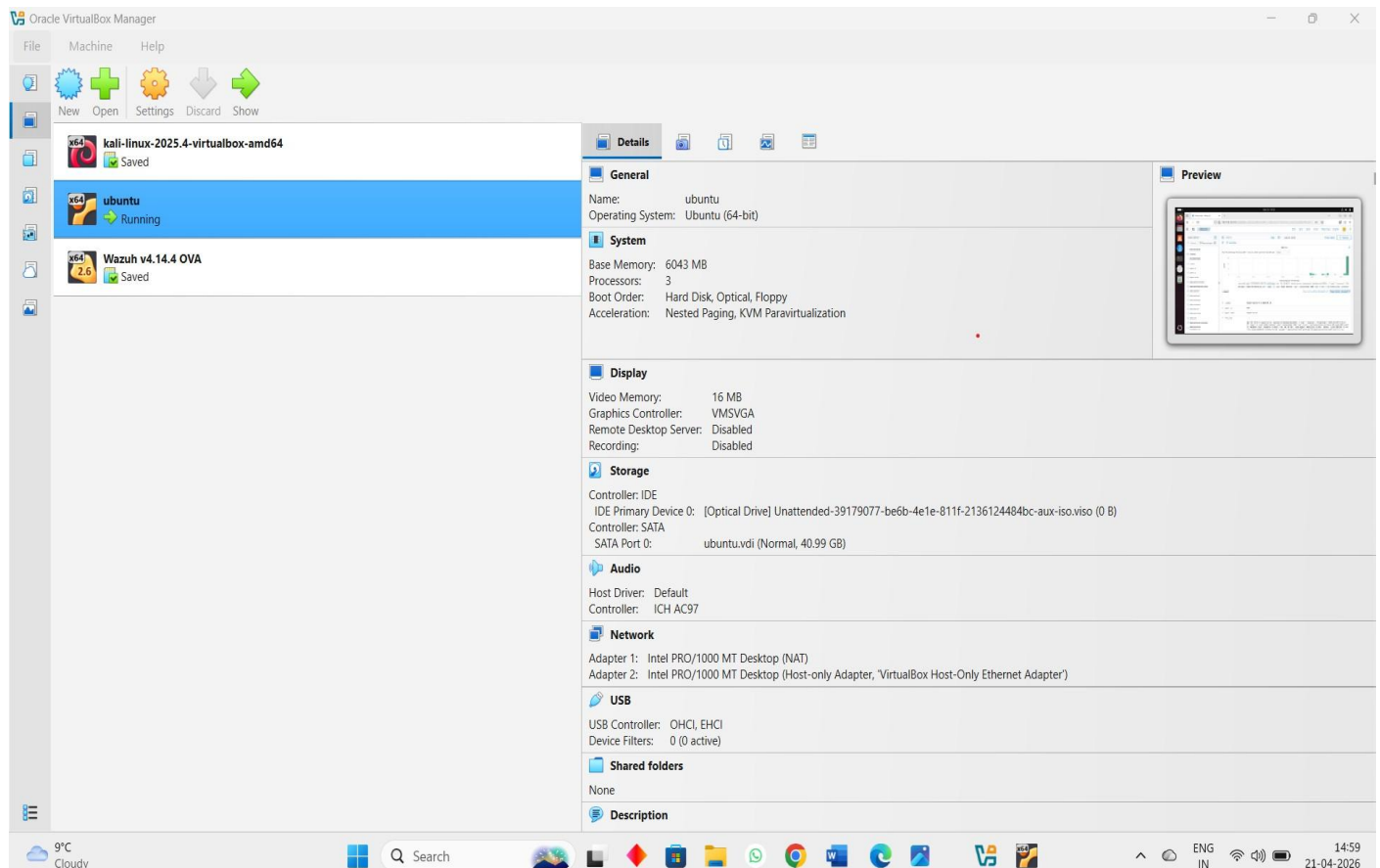


Figure 1: Virtual Machine Setup In Virtualbox (Kali Linux, Ubuntu, Wazuh)

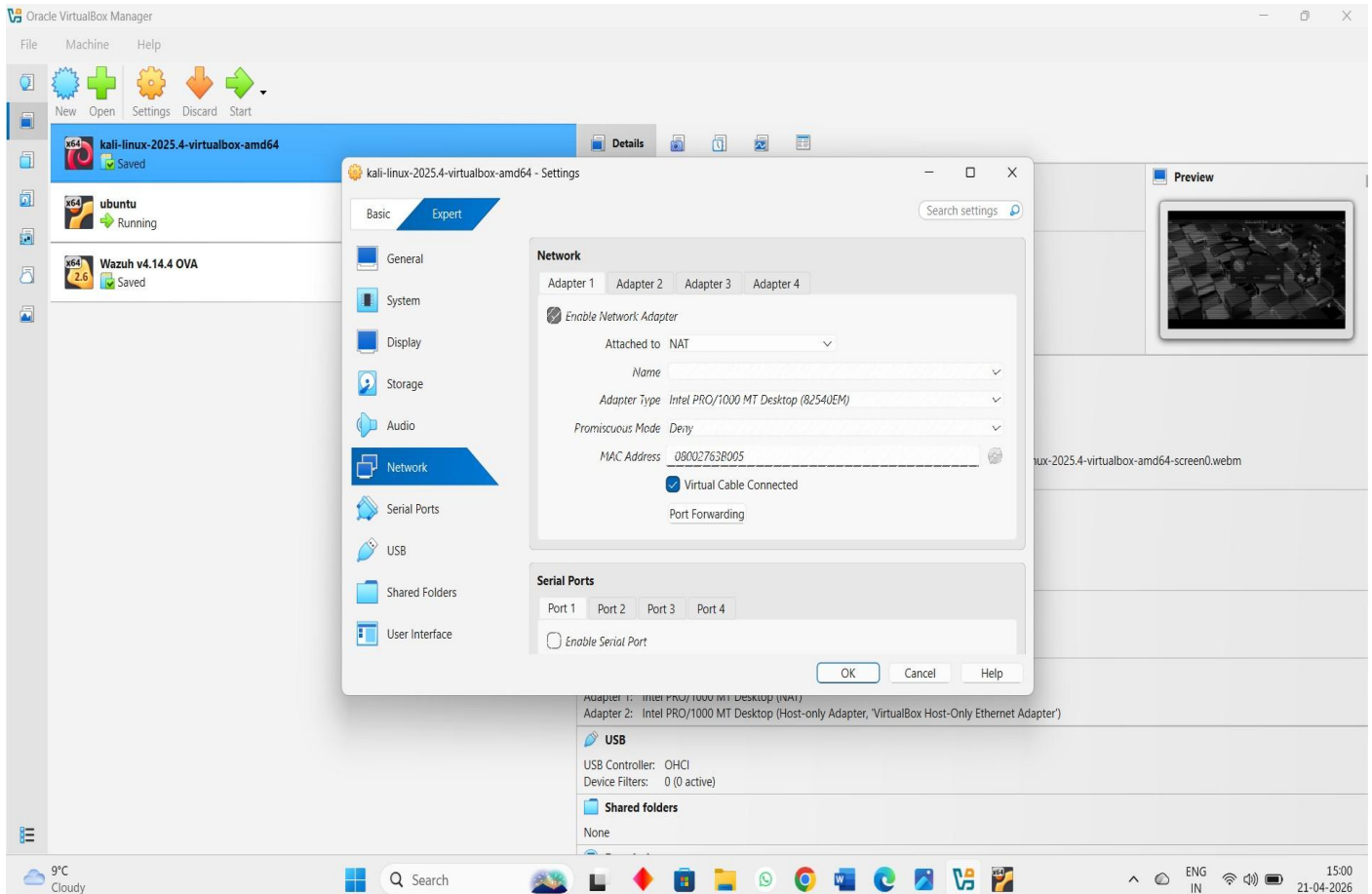


Figure 2: NAT Network Configuration

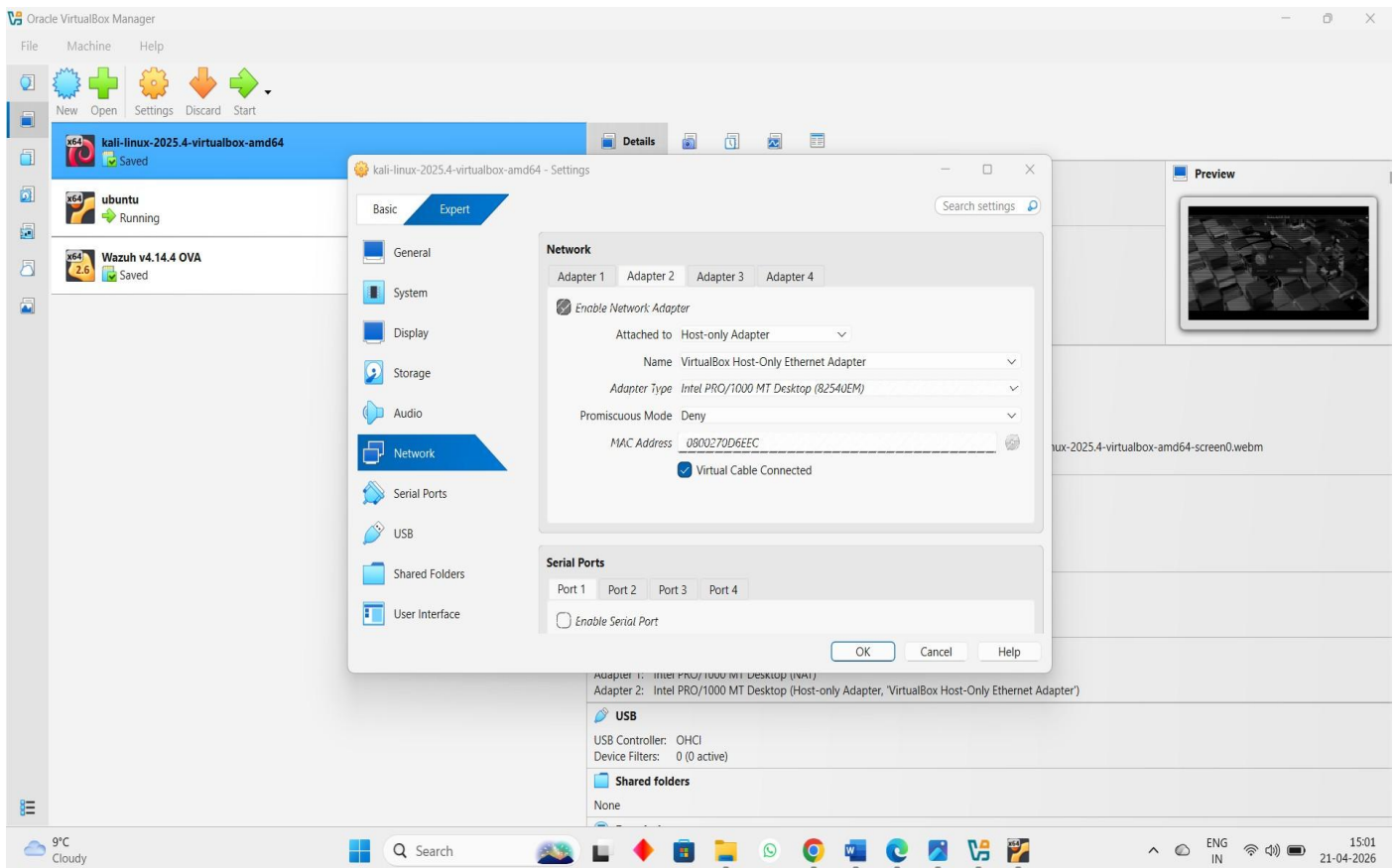


Figure 3: Host-Only Network Configuration

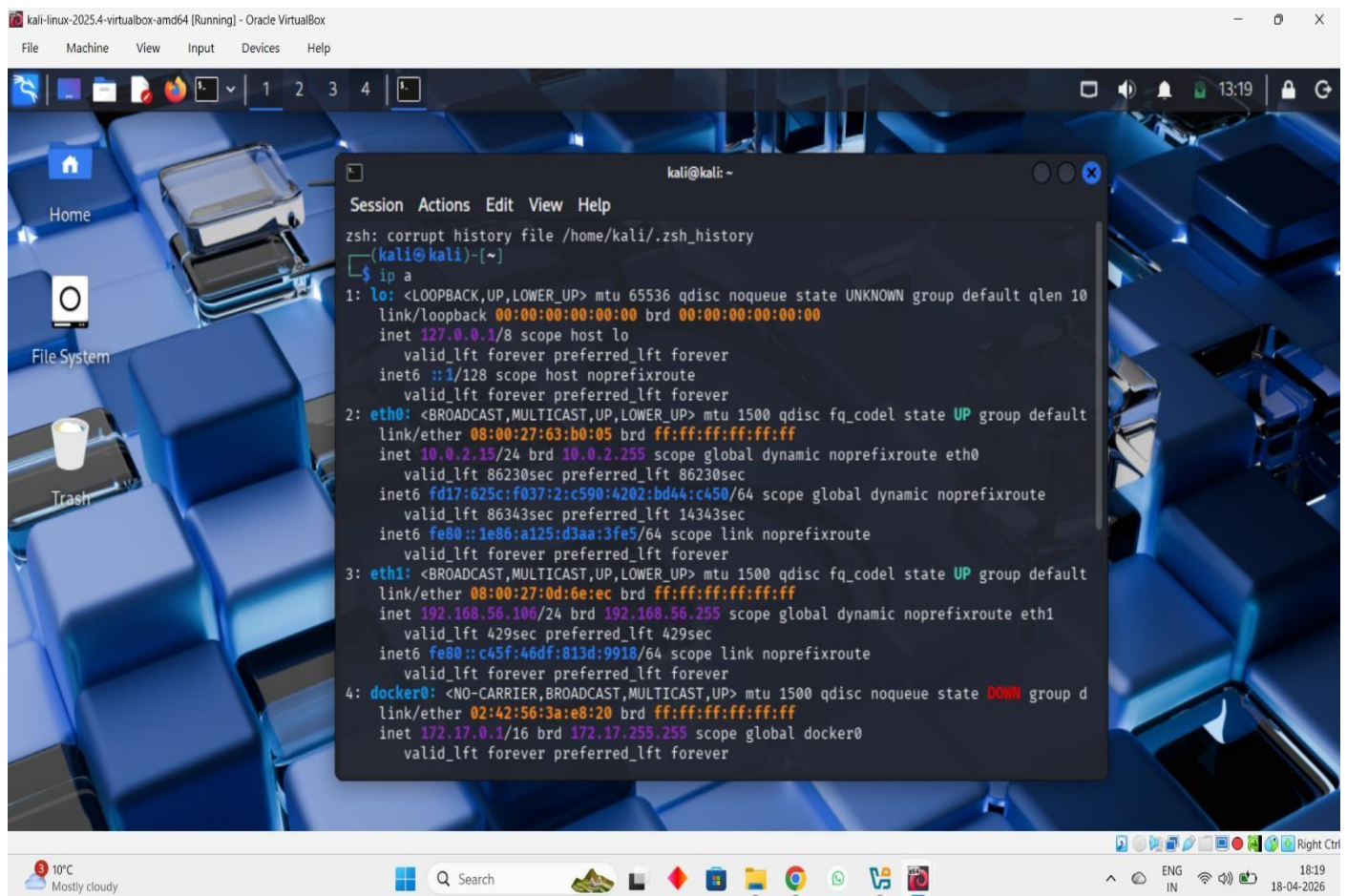


Figure 4: Ip Address Configuration Of Kali Linux (192.168.56.106)

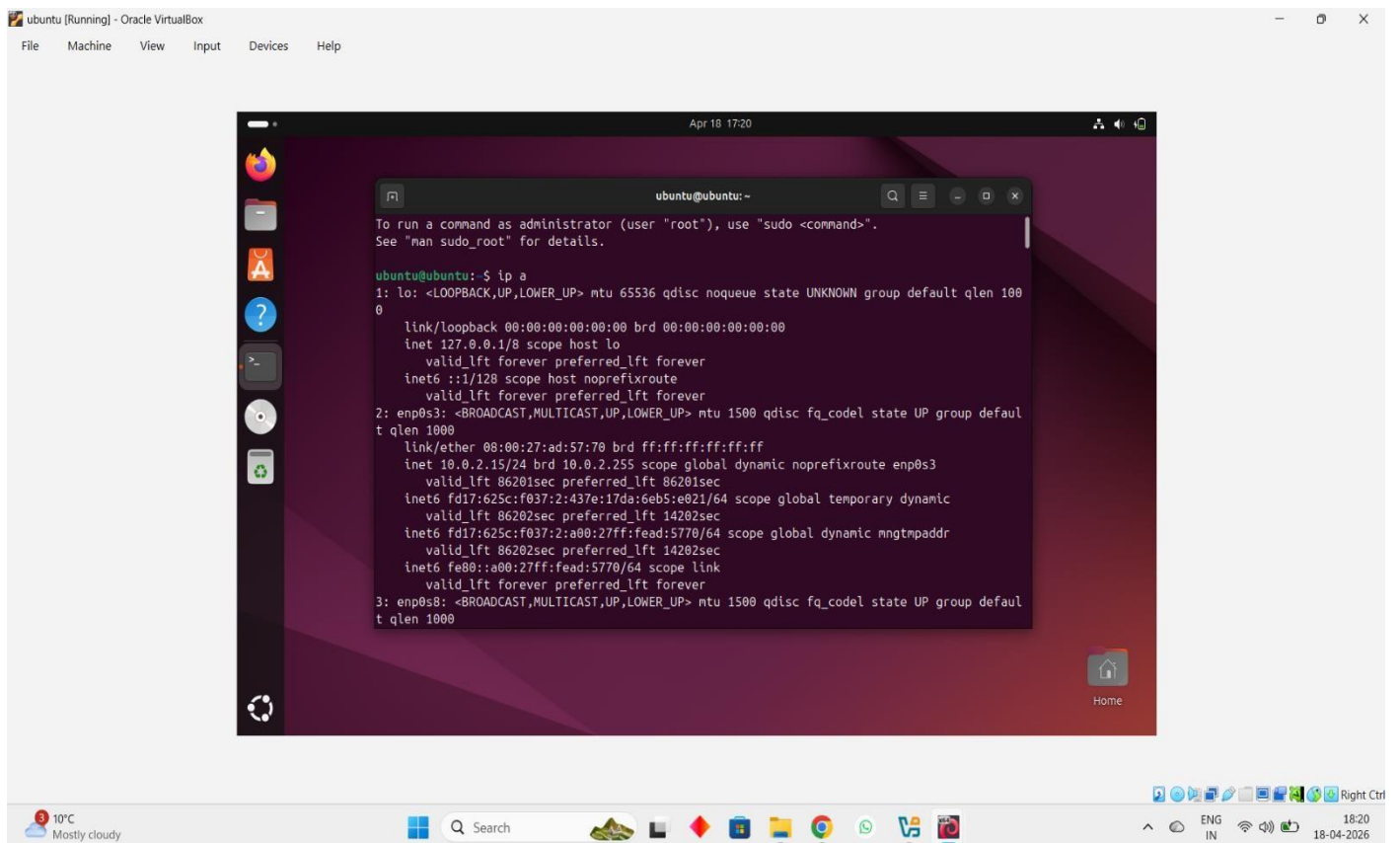


Figure 5: IP Address Configuration Of Ubuntu (192.168.56.102)

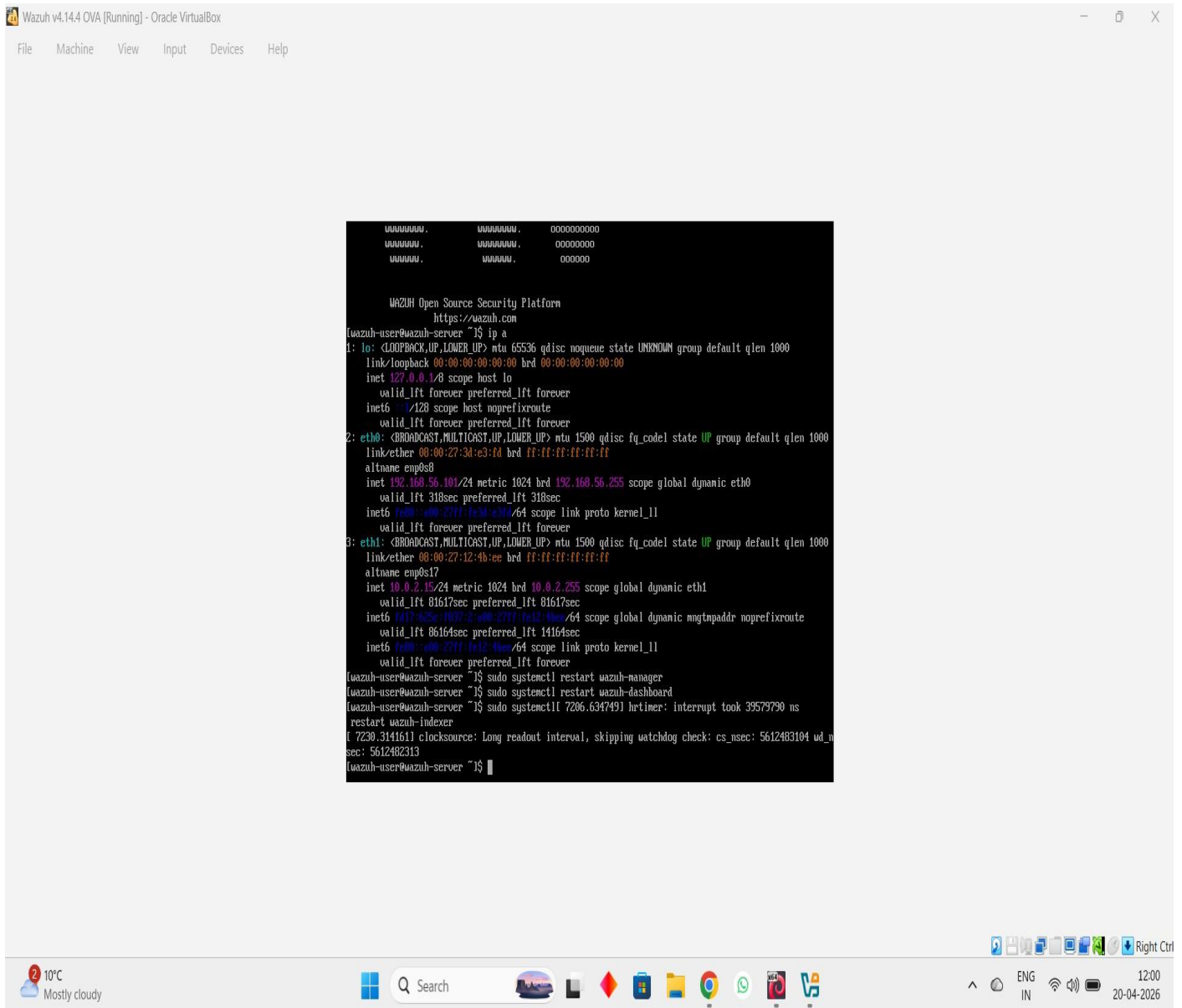


Figure 6: IP Address Configuration of Wazuh Server (192.168.56.101)

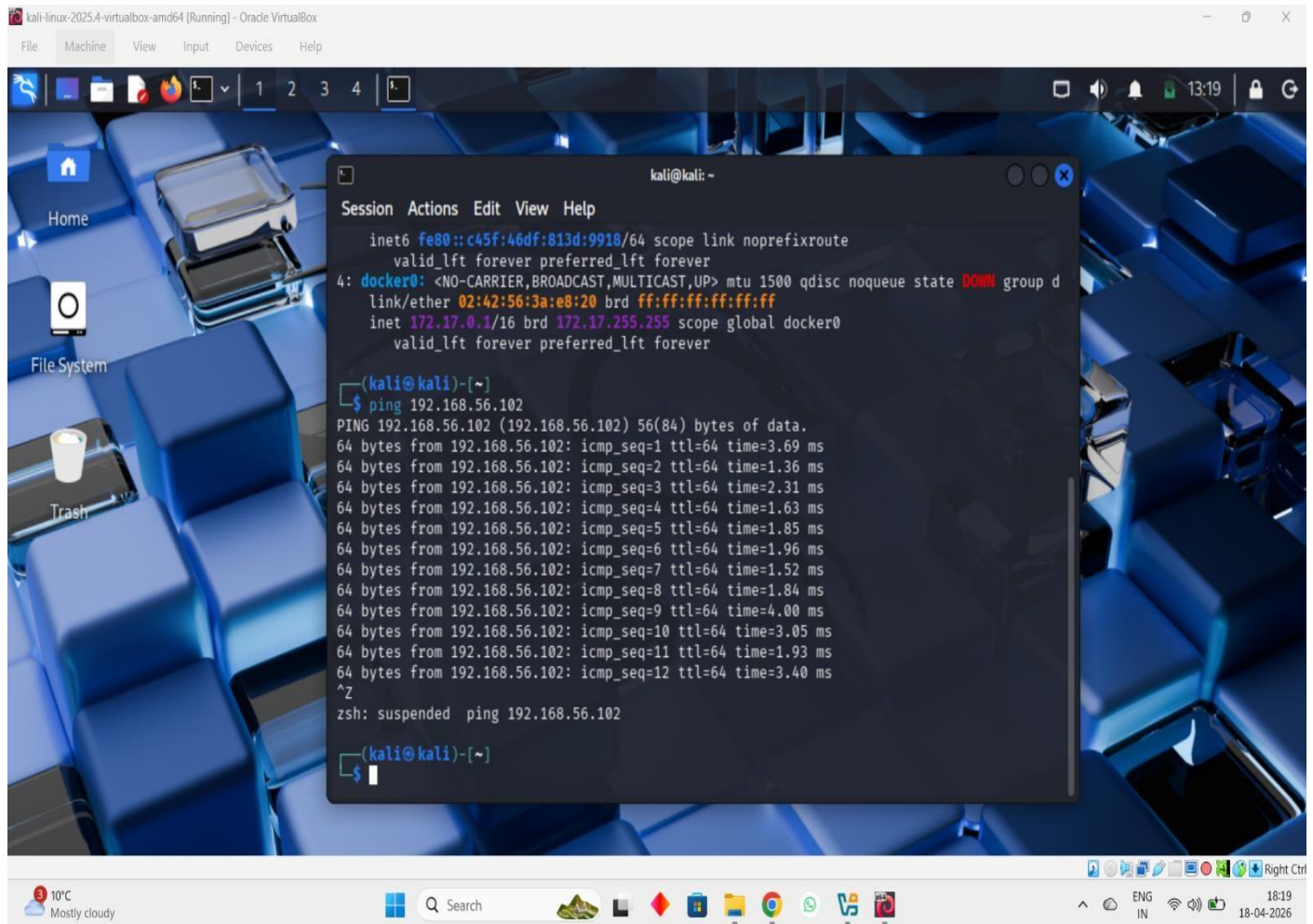


Figure 7: Network Connectivity Test – Ping From Kali To Ubuntu

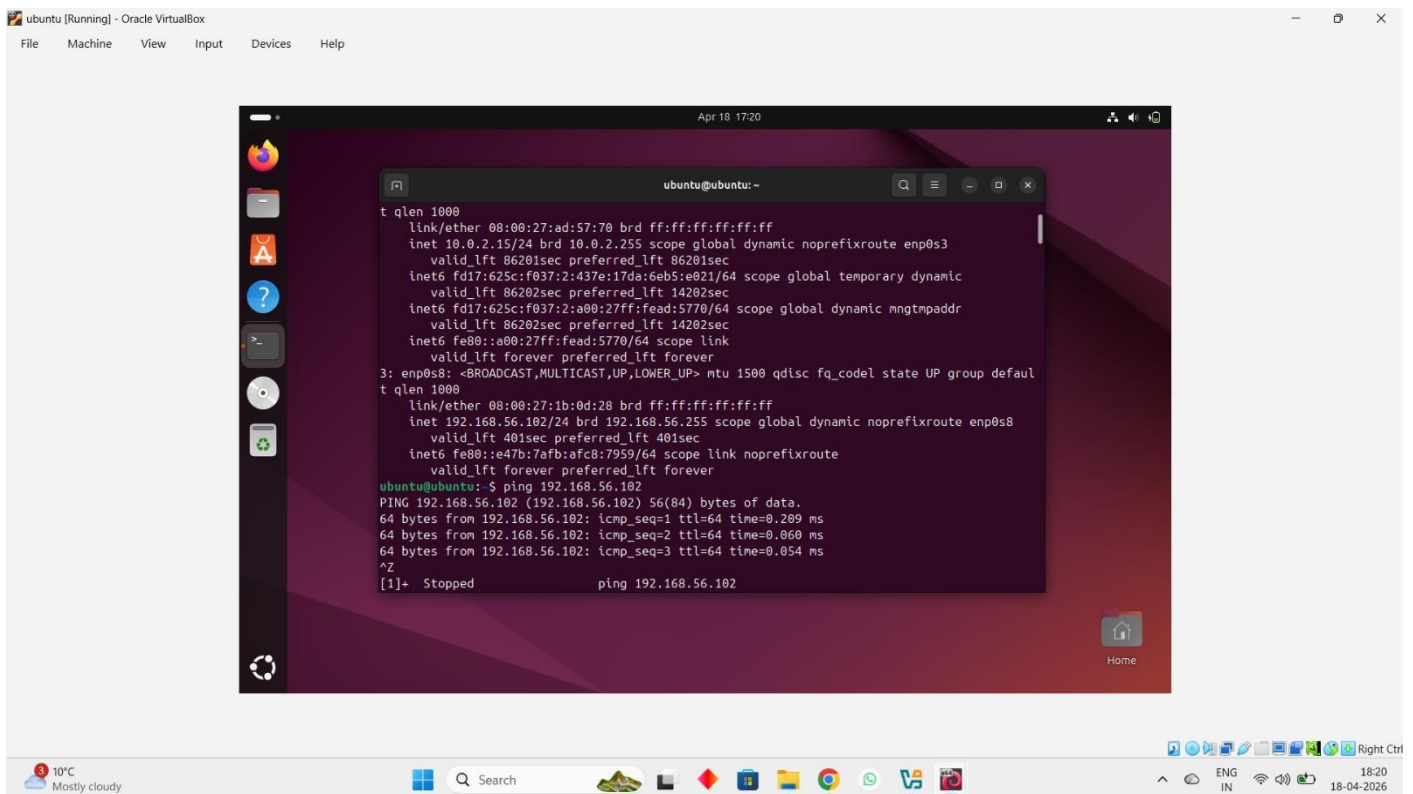


Figure 8: Network Connectivity Test – Ping from Ubuntu to Kali

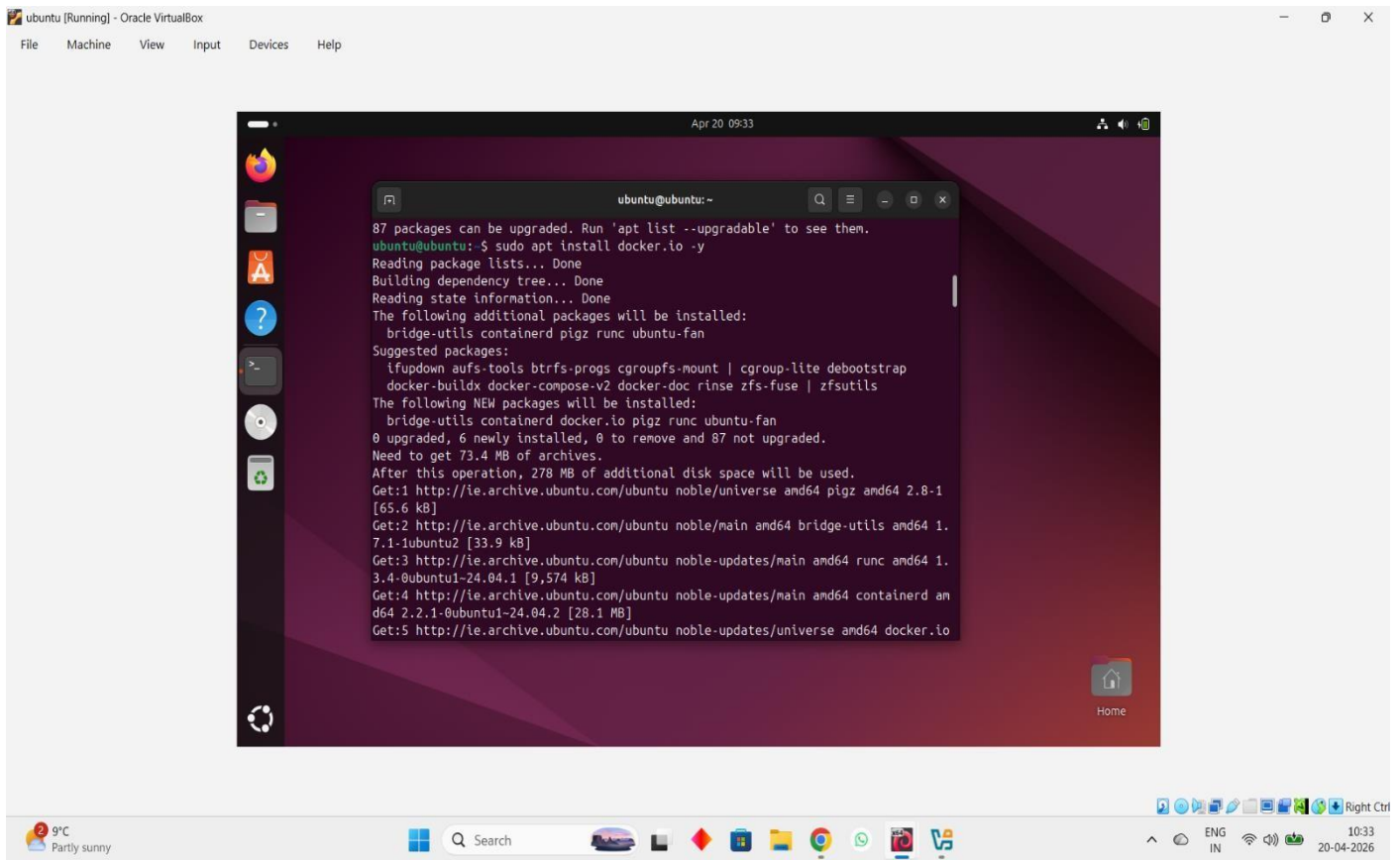


Figure 9: Docker Installation and Configuration on Ubuntu

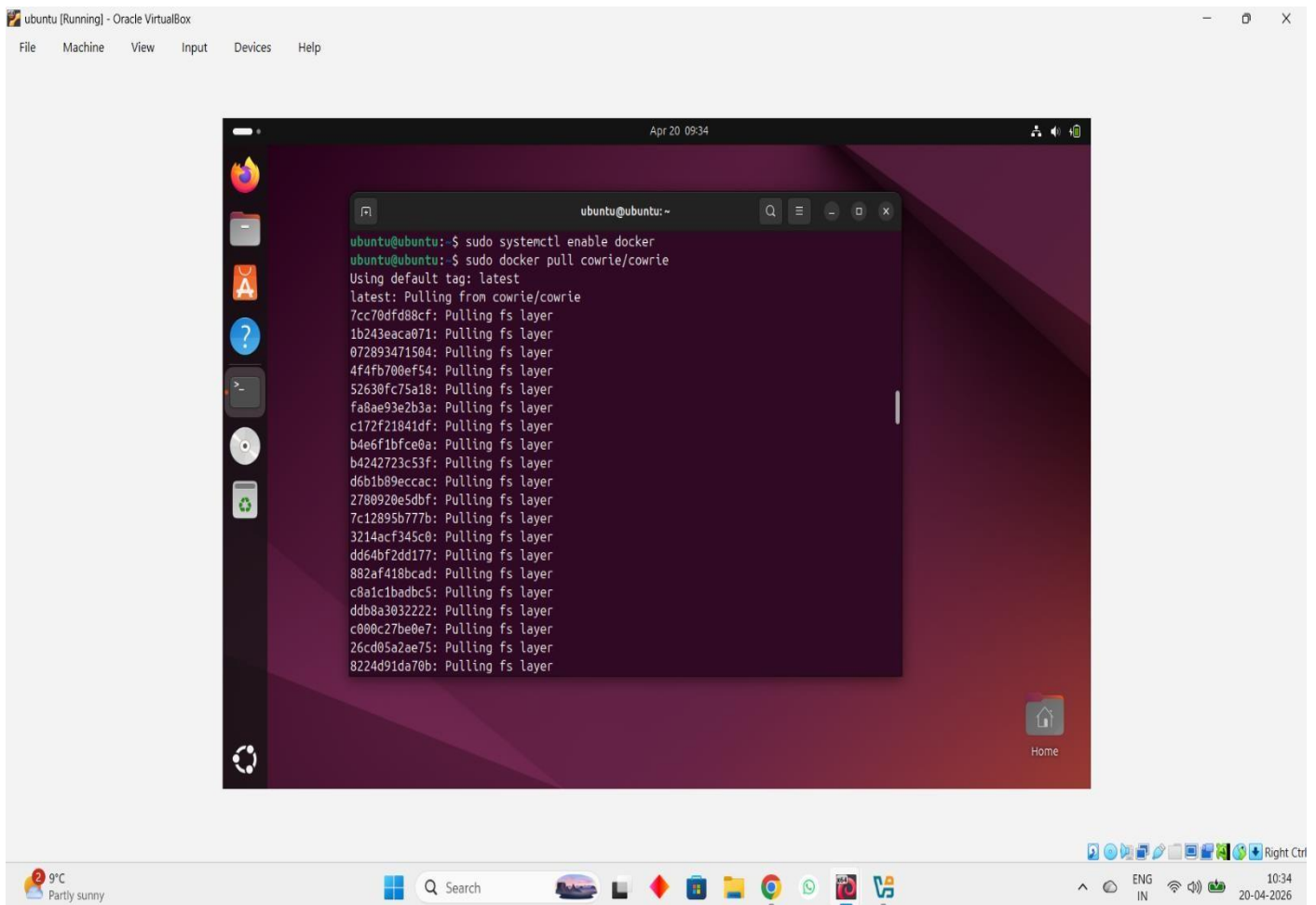


Figure 10: Pulling Cowrie Image using Docker

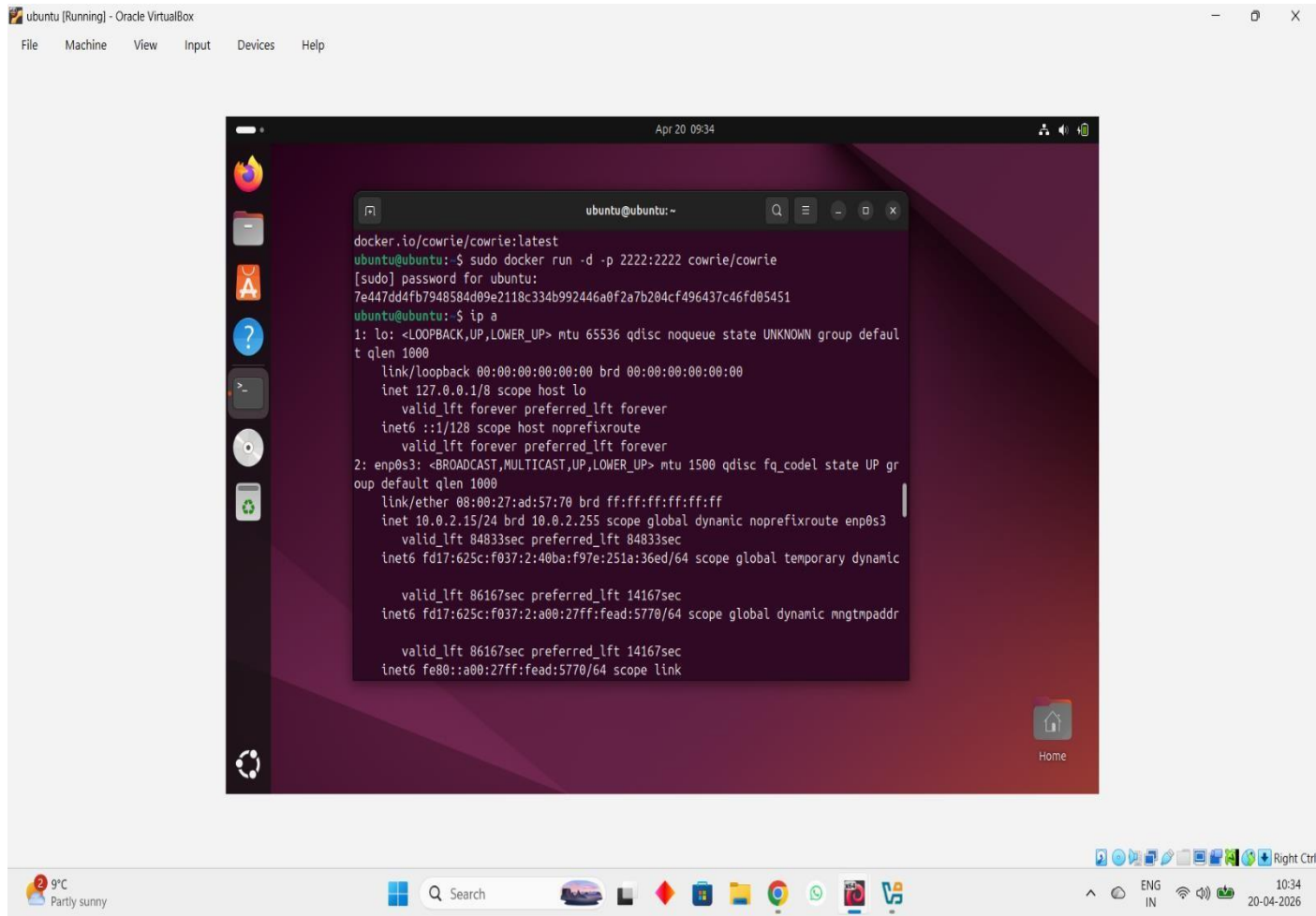


Figure 11: Running Cowrie Honeypot using Docker

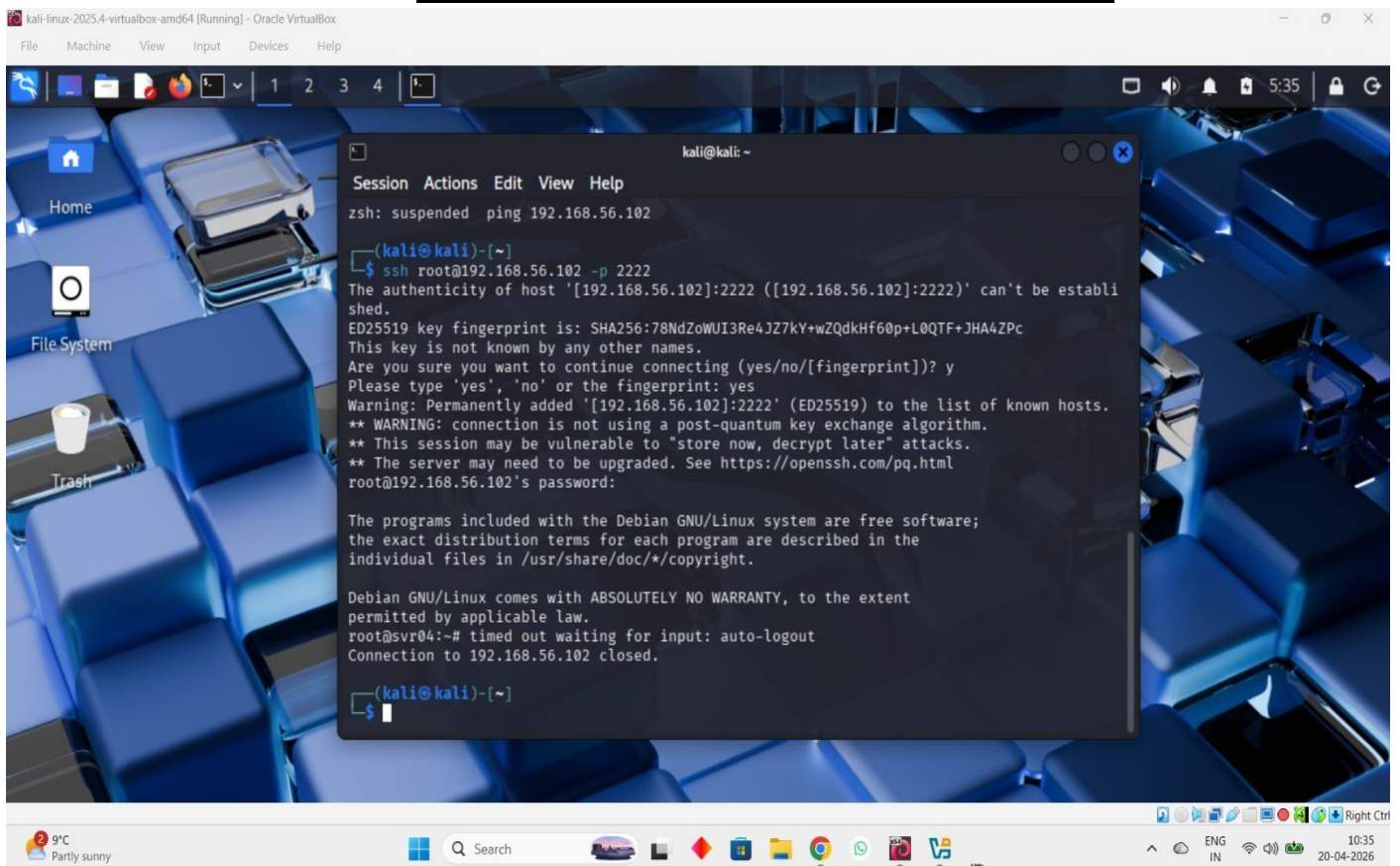


Figure 12: SSH Attack Simulation from Kali Linux

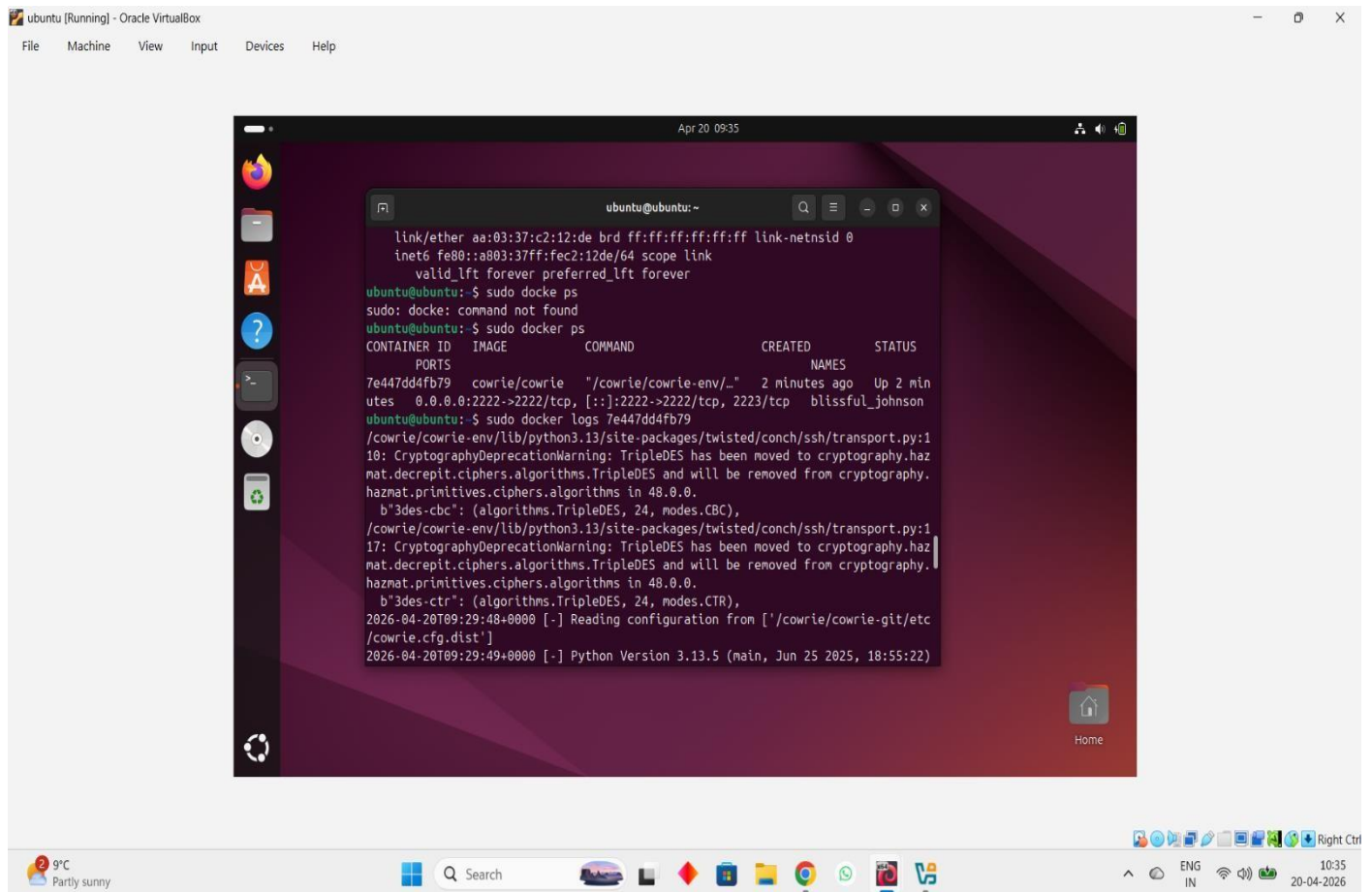


Figure 13: Viewing Cowrie Logs using Docker on Ubuntu

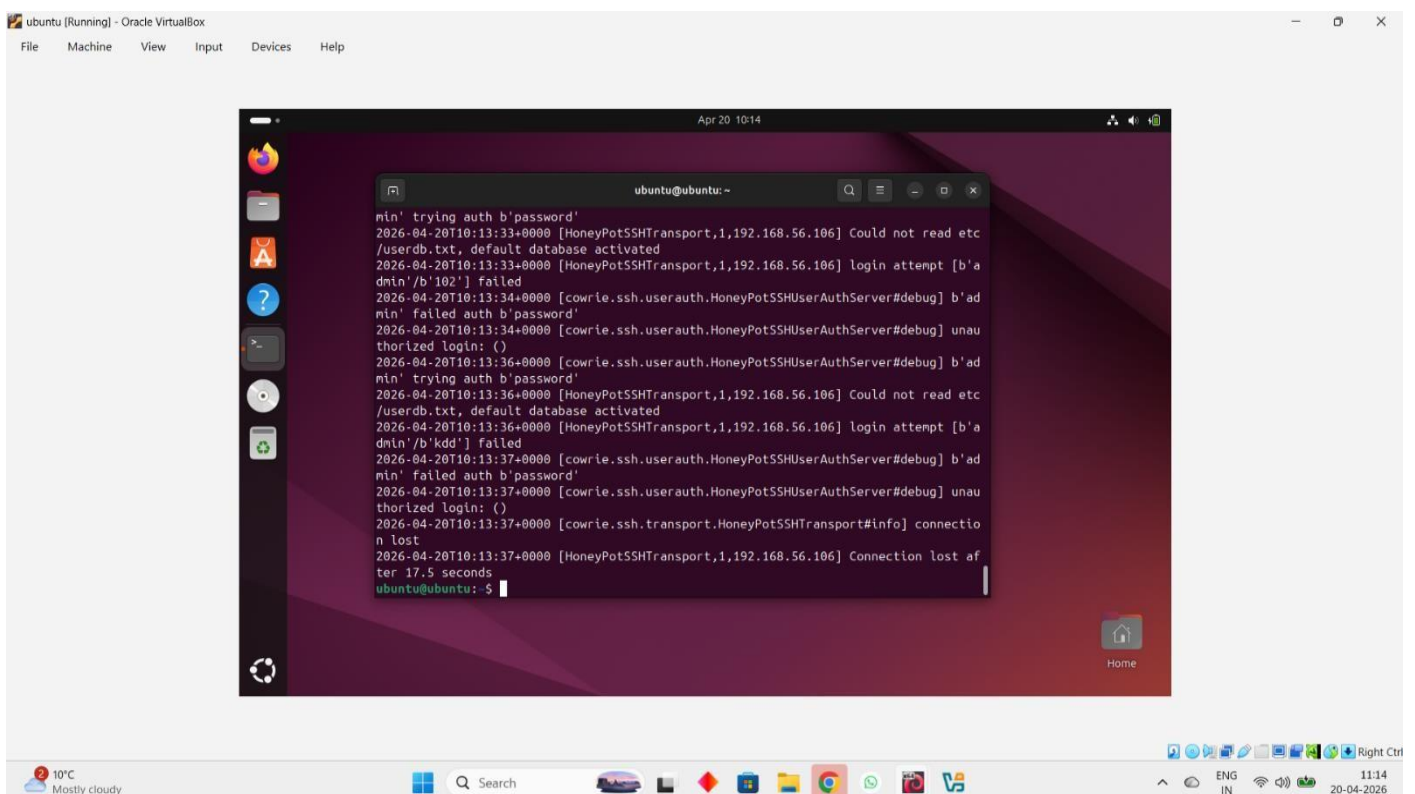


Figure 14: Cowrie Log Output Showing Attacker Activity

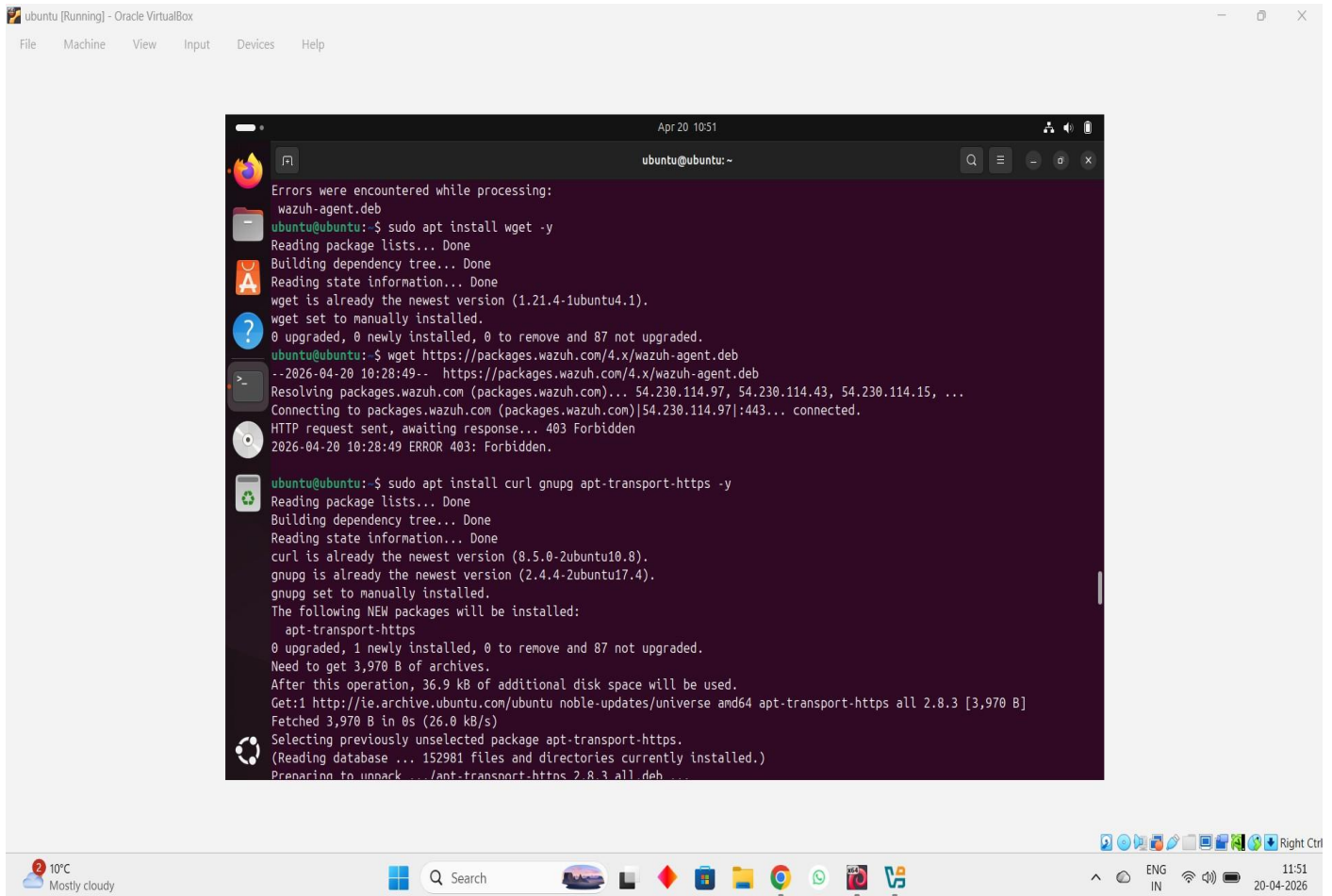


Figure 15: Installation Of Wget On Ubuntu

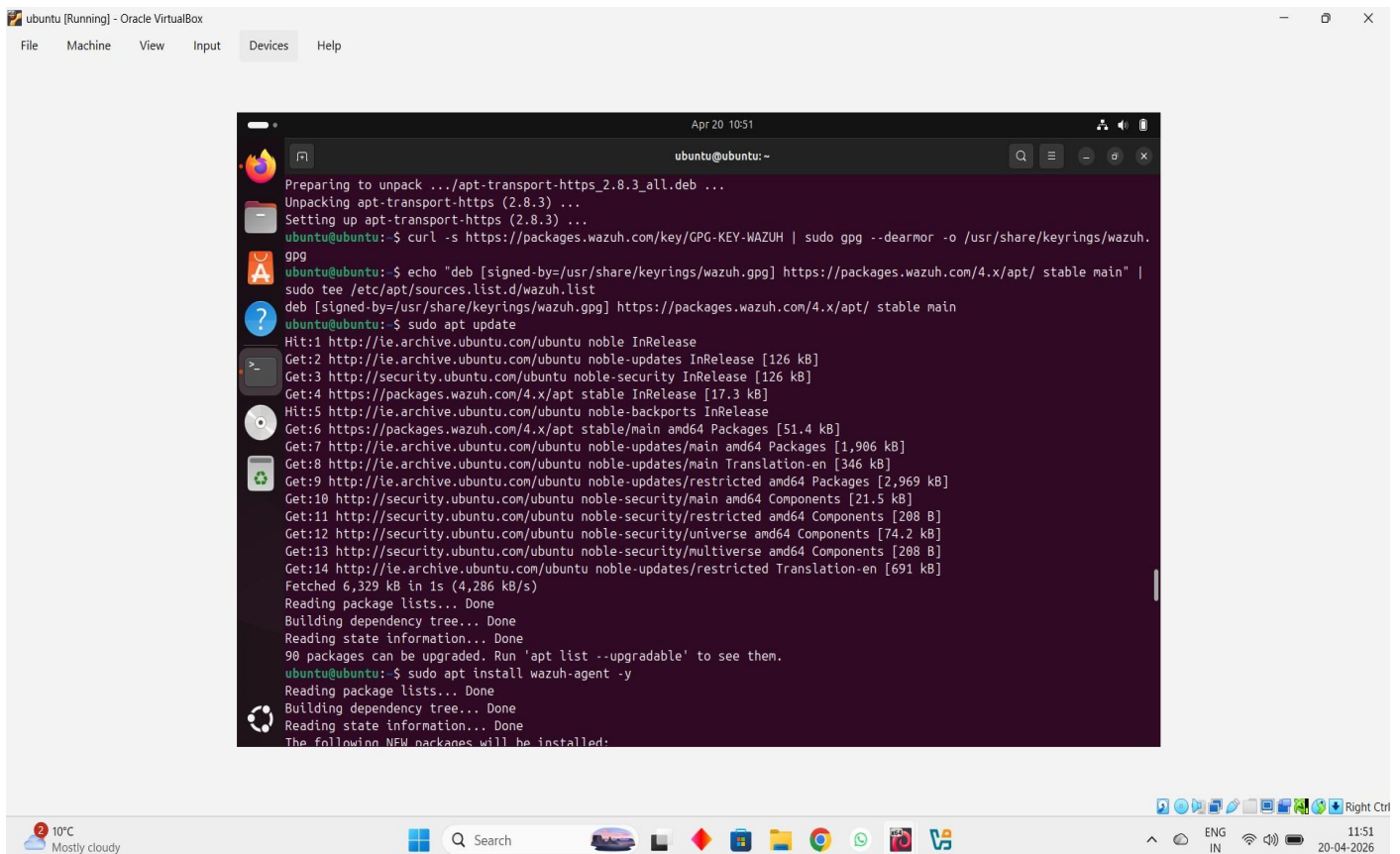


Figure 16: Updating System And Installing Wazuh Agent

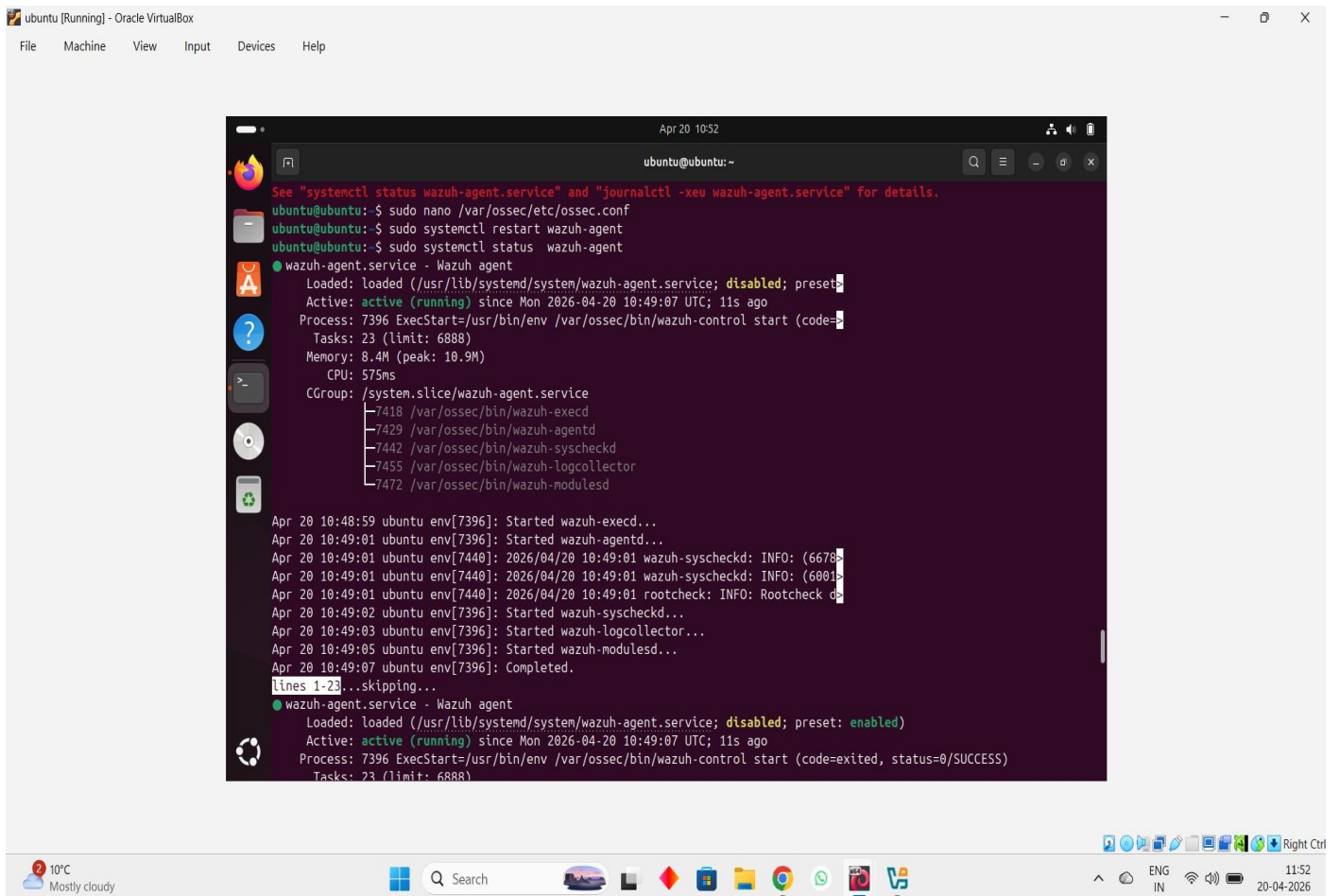


Figure 17: Restarting And Checking Status Of Wazuh Agent

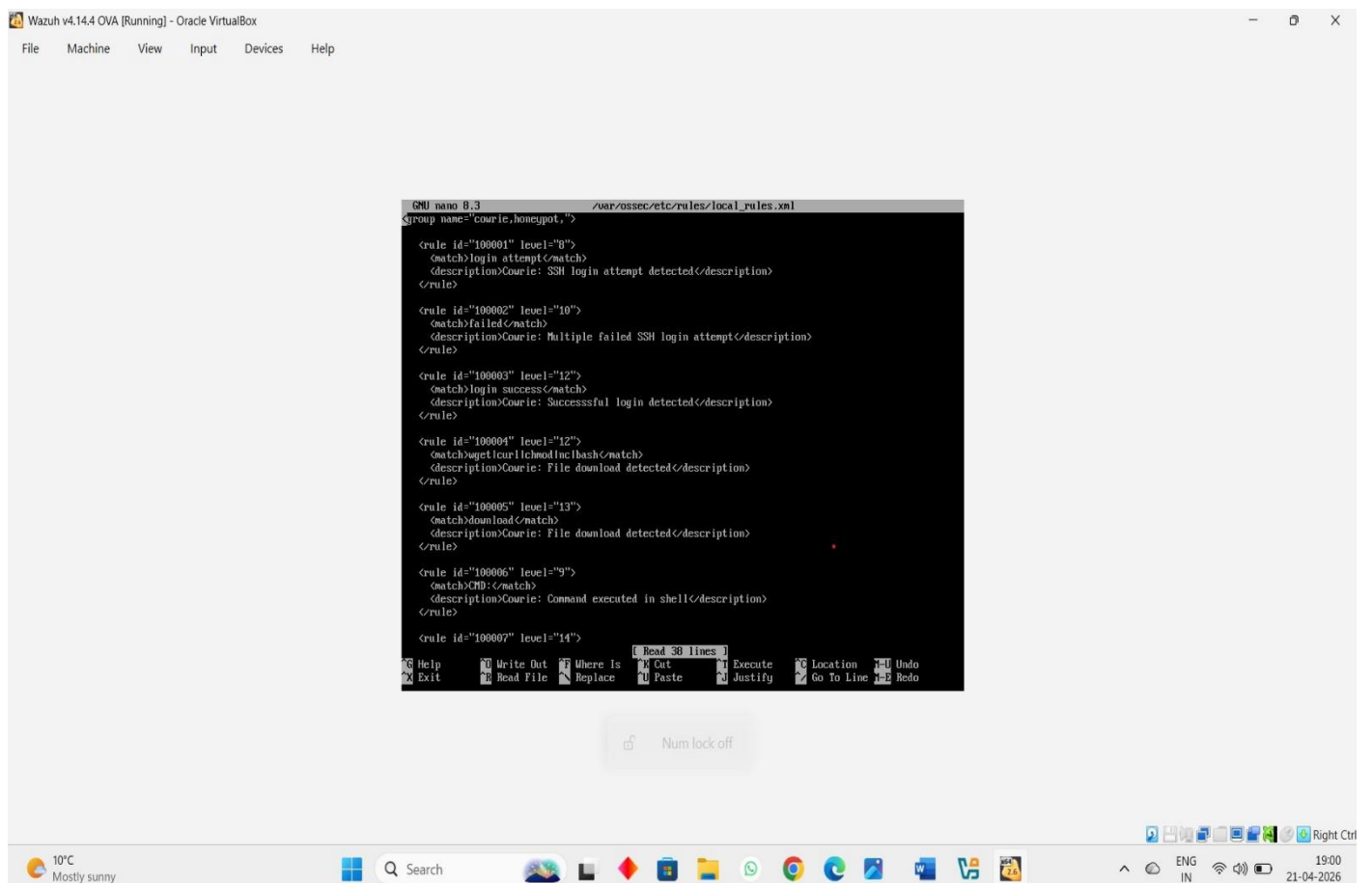


Figure 18: Custom Detection Rules Created In Wazuh

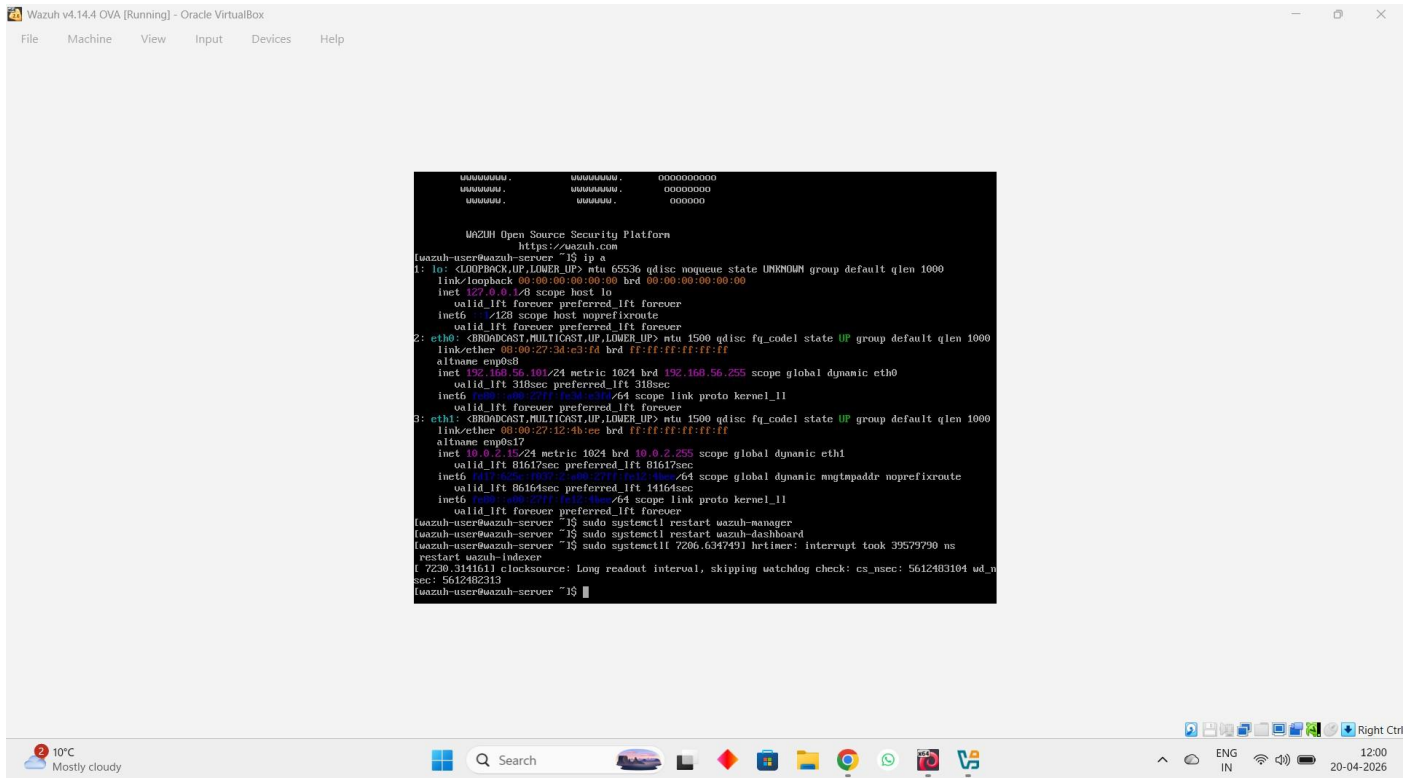


Figure 19: Restarting Wazuh Services (Manager, Dashboard, Indexer)

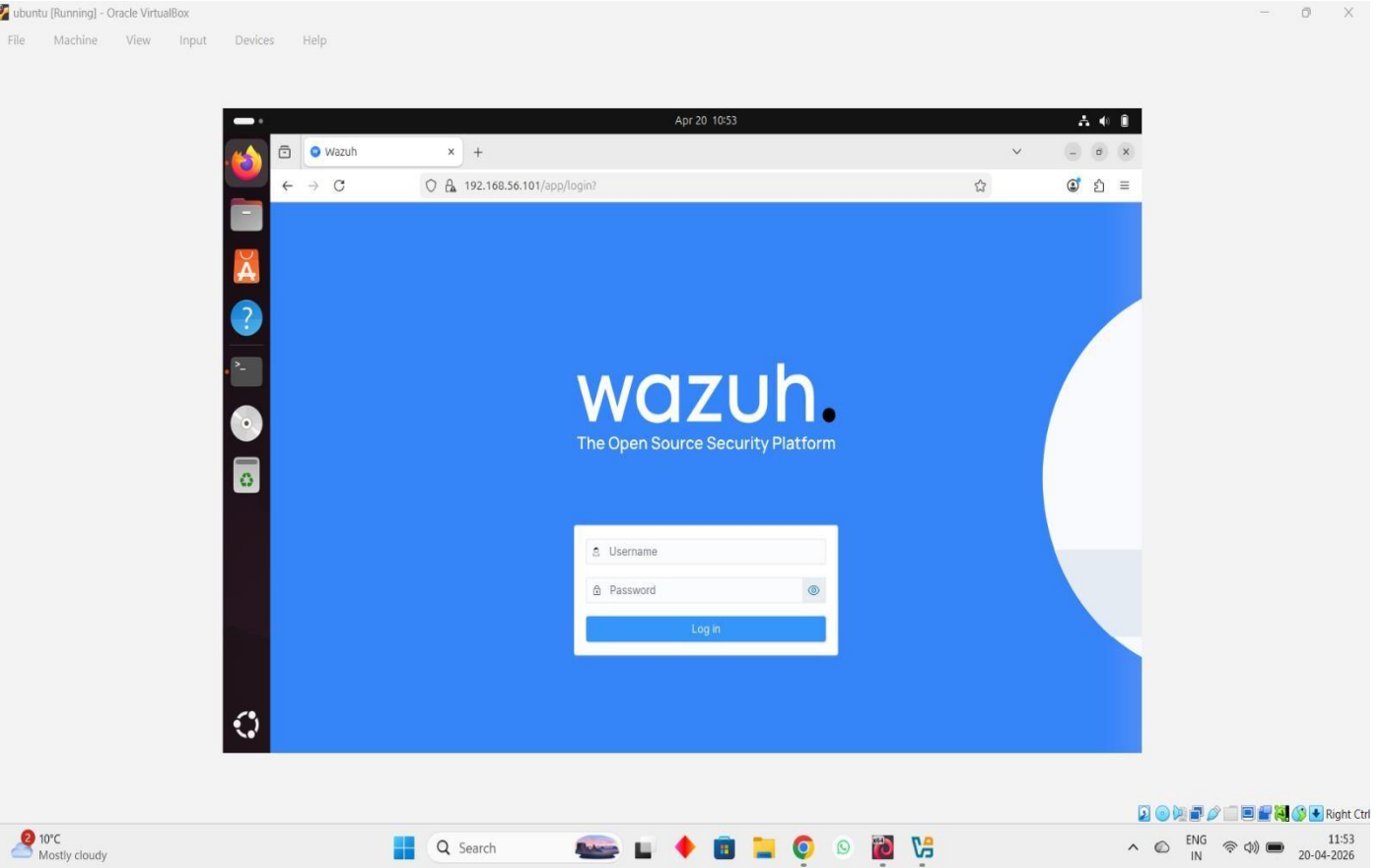


Figure 20: Accessing Wazuh Dashboard in Web Browser

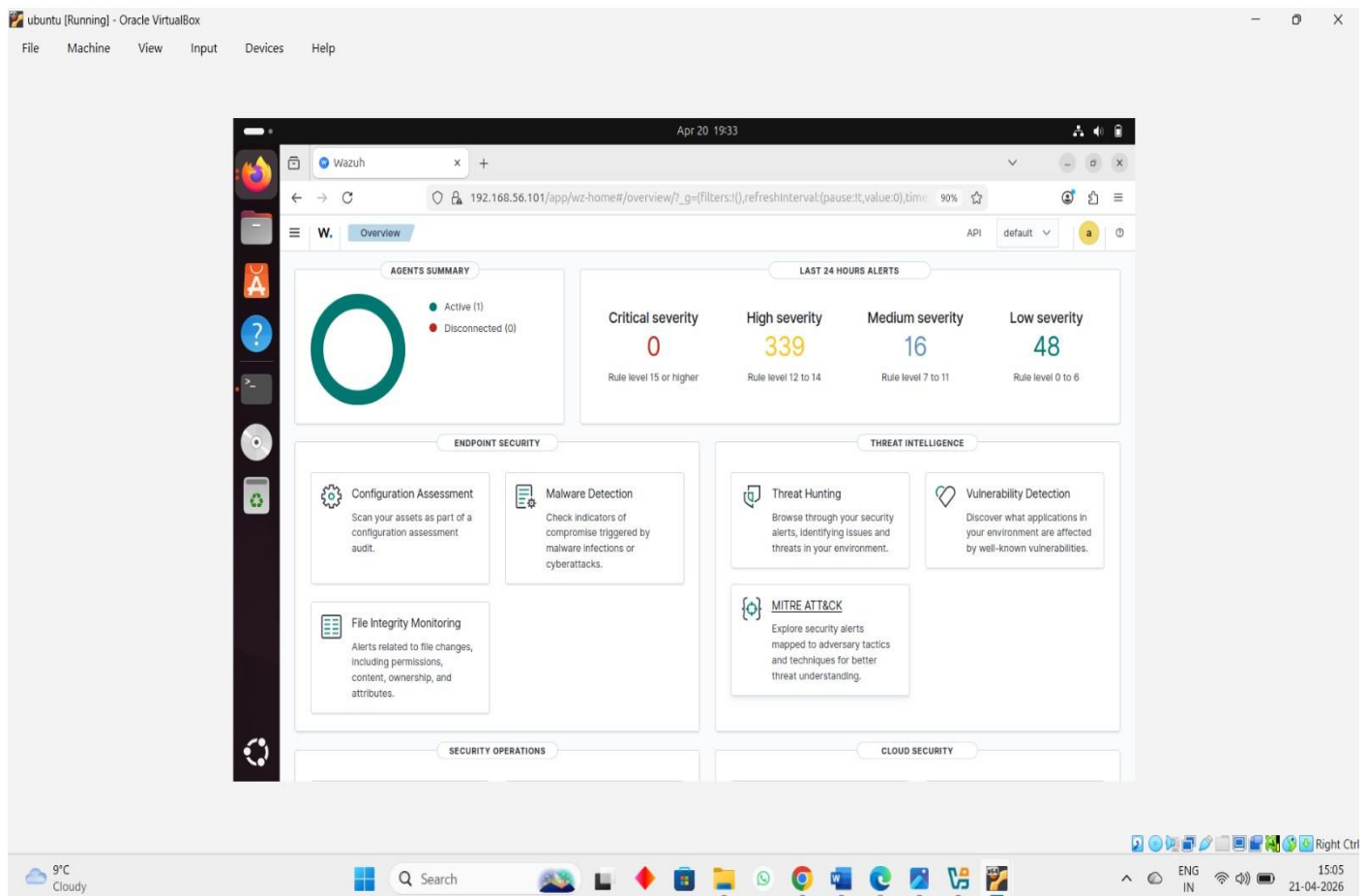


Figure 21: Wazuh Dashboard Showing Active Agent (Ubuntu)

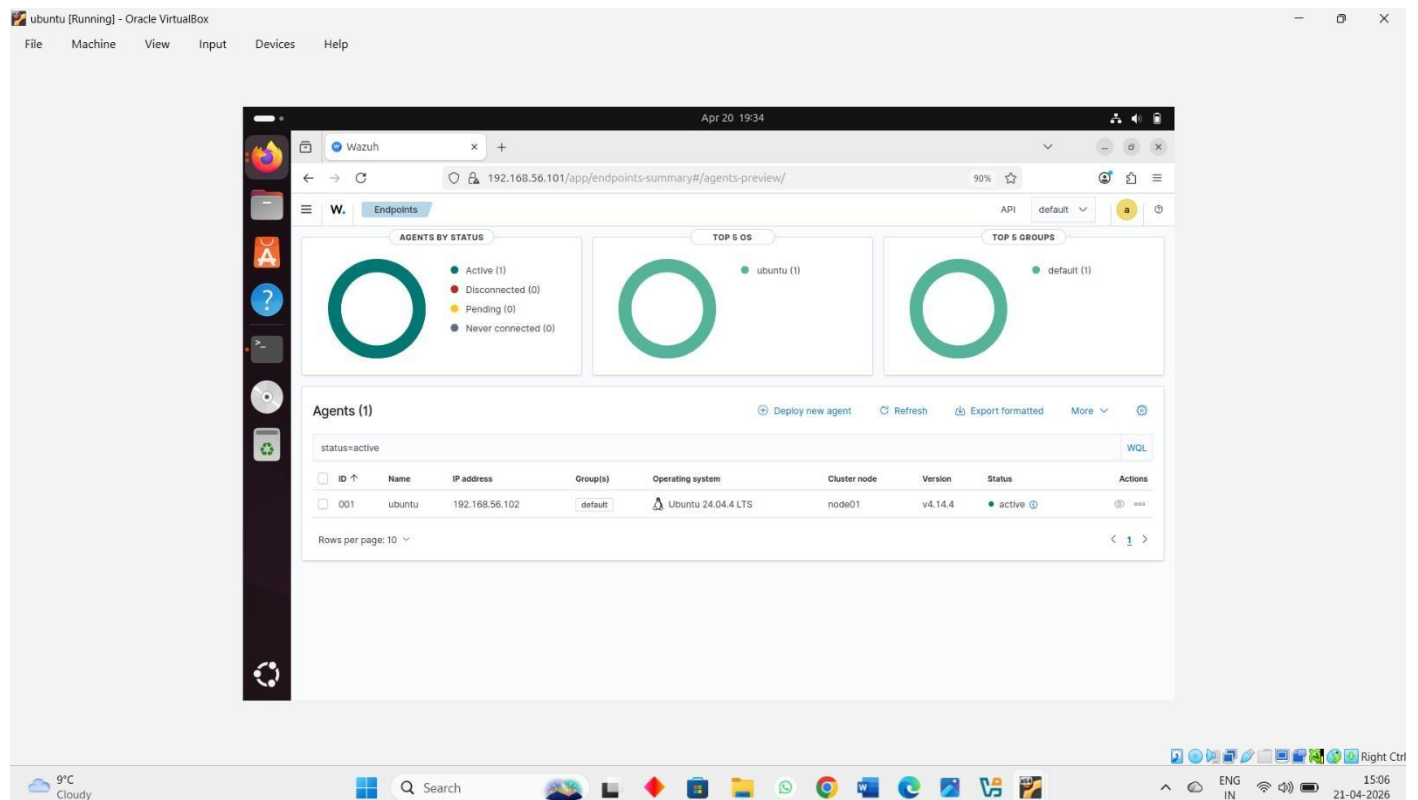


Figure 22: Wazuh Dashboard Showing Active Endpoint (Cowrie Honeypot)

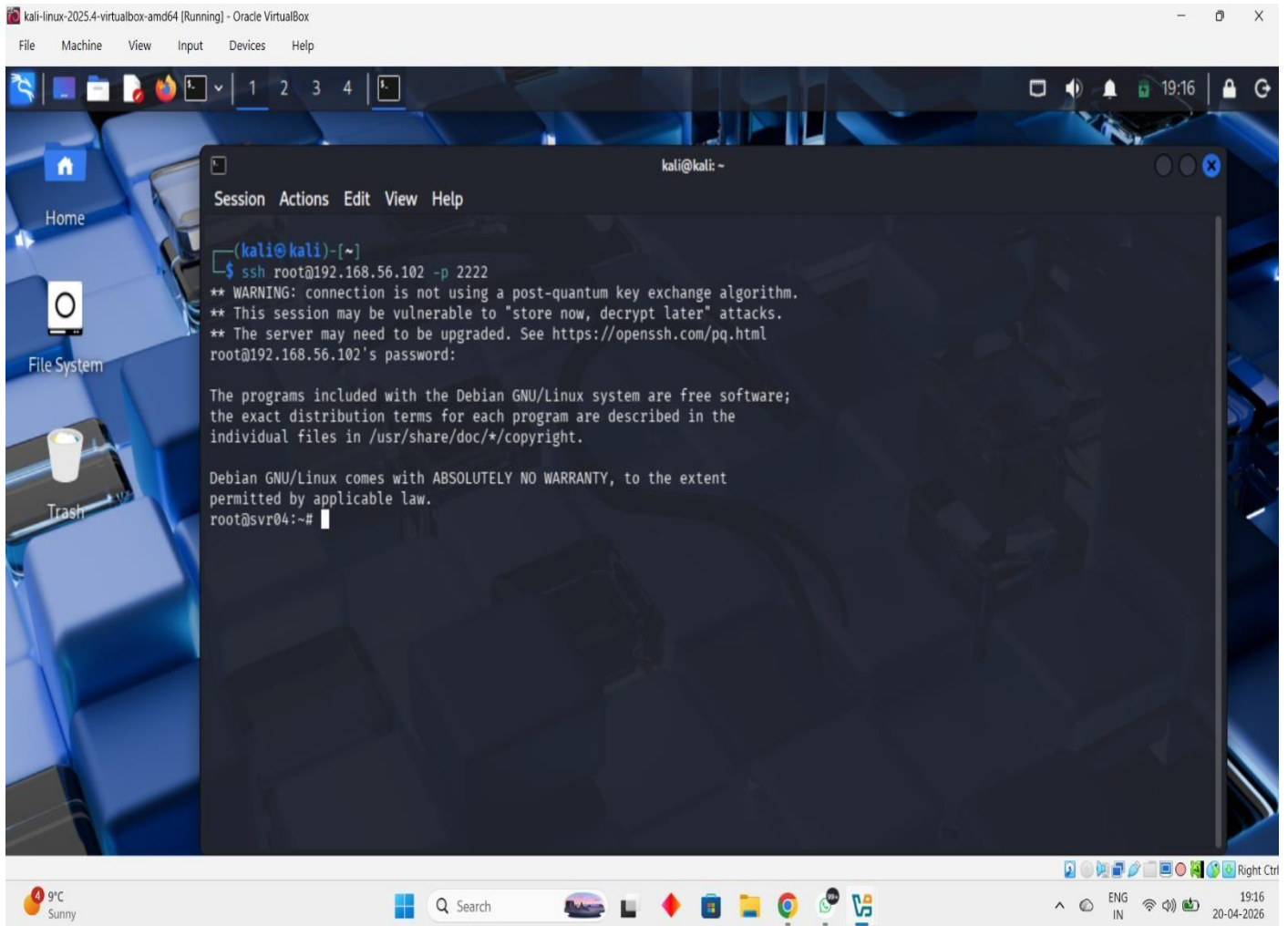


Figure 23: SSH Attack Testing From Kali Linux

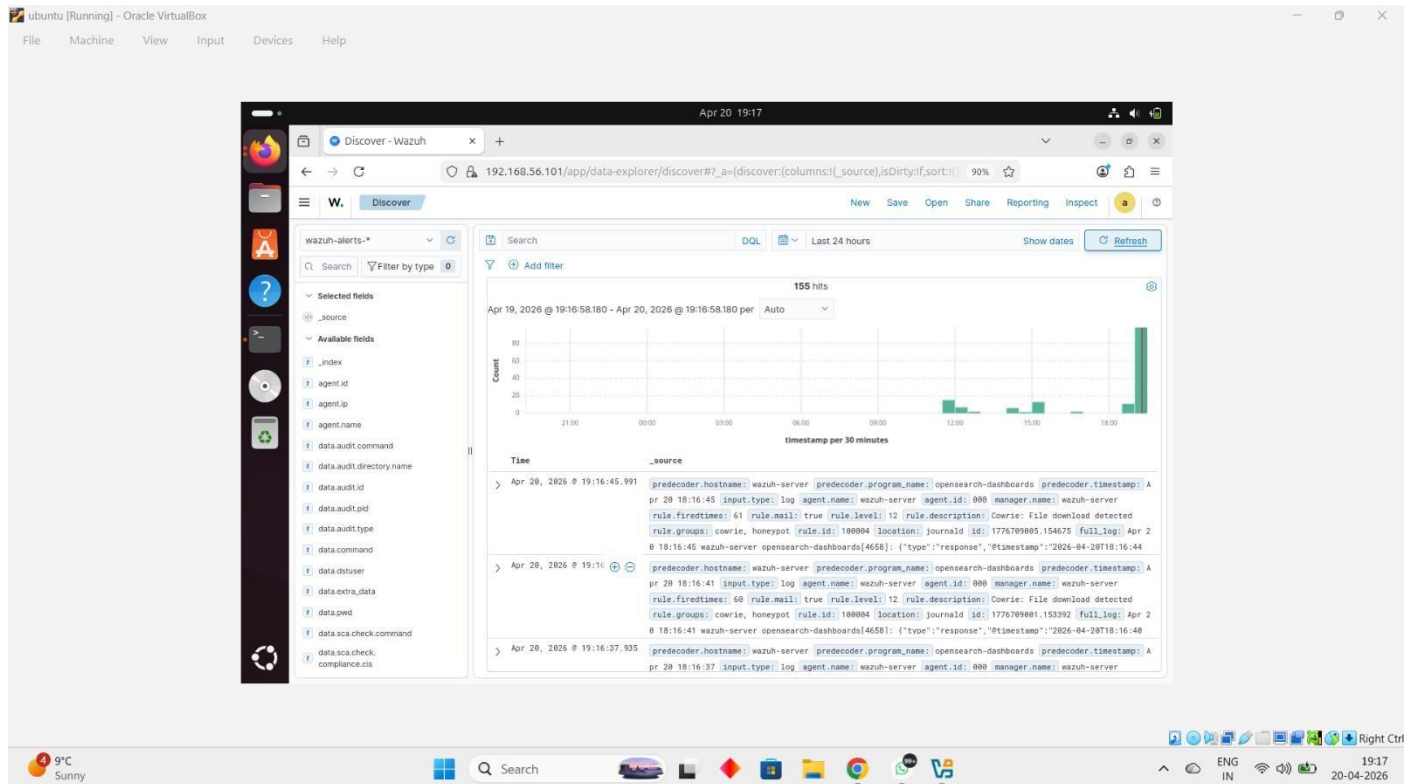


Figure 24: Viewing Logs In Wazuh Discover Section

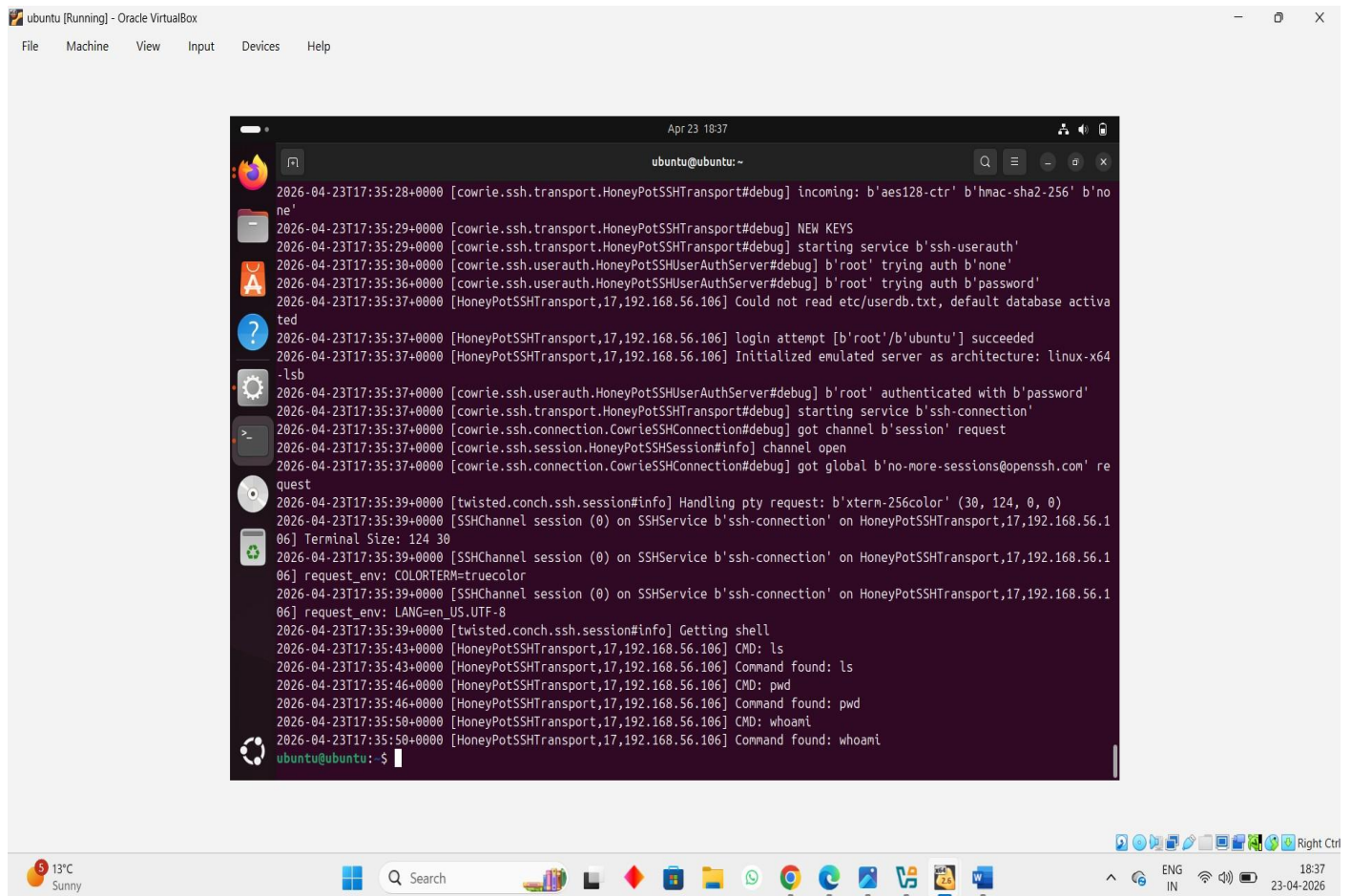


Figure 27: Cowrie Logs Capturing Attacker Activity on Ubuntu

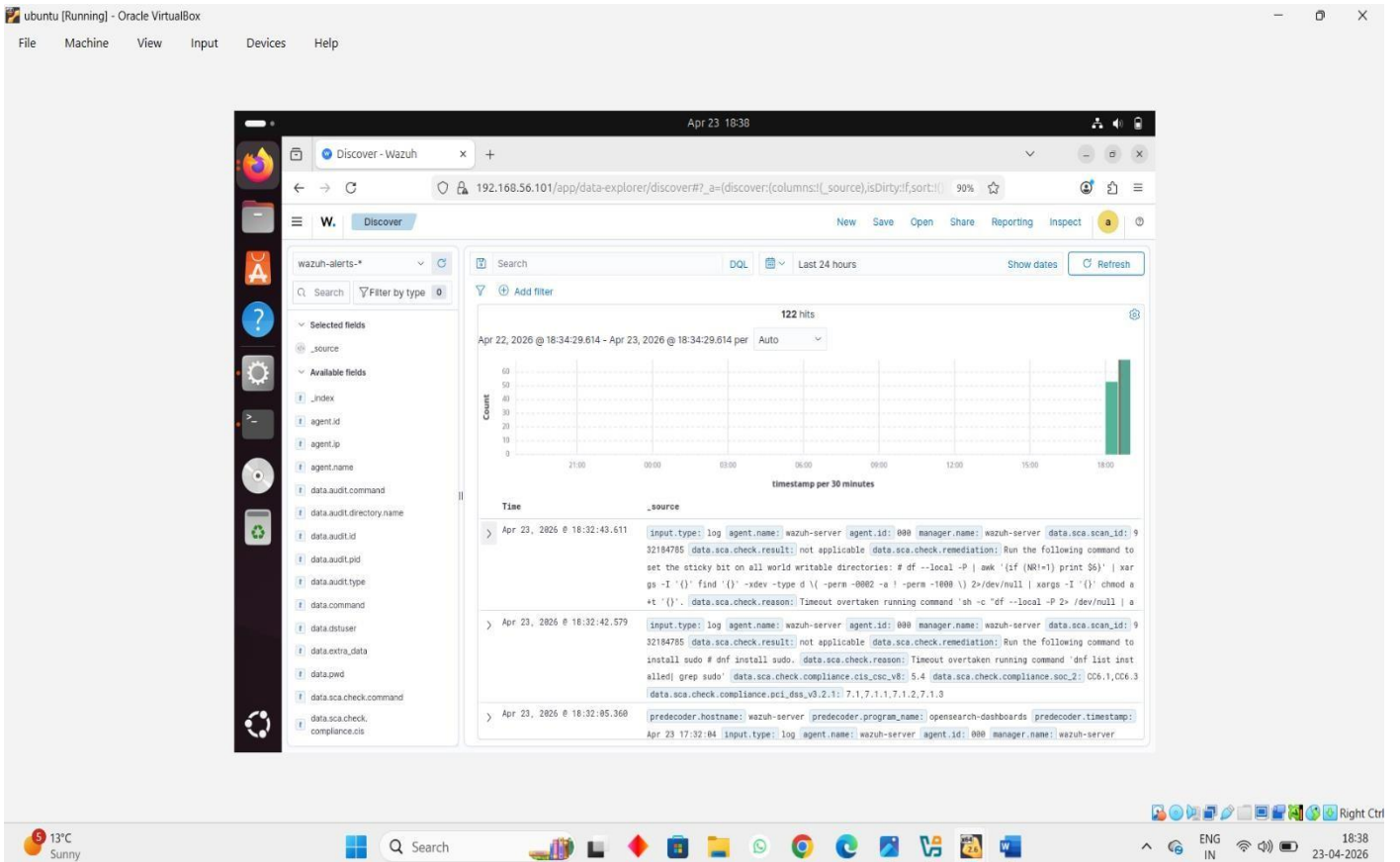
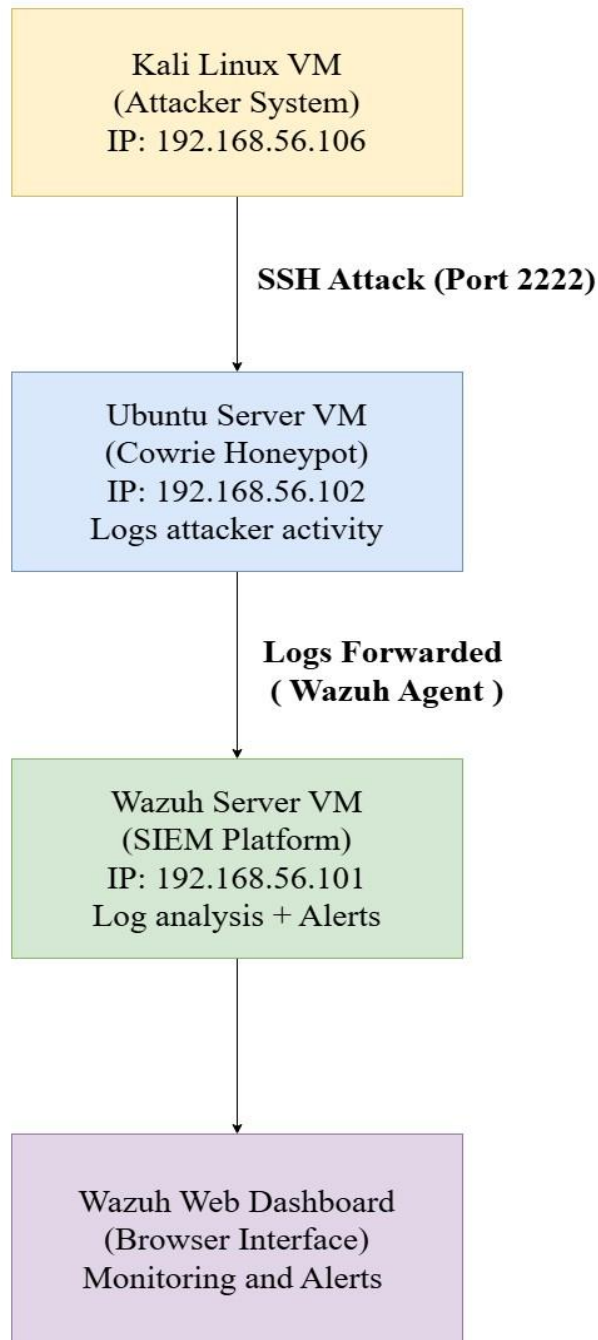


Figure 28: Attacker Activity Visualized in Wazuh Dashboard

ARCHITECTURE DIAGRAM



**Network: Host-Only (192.168.56.x) + NAT
(Internet Access)**

Figure 29: Architecture Of Honeypot Lab Environment

VIDEO LINK :- <https://youtu.be/ktG1NvDb9Ug>